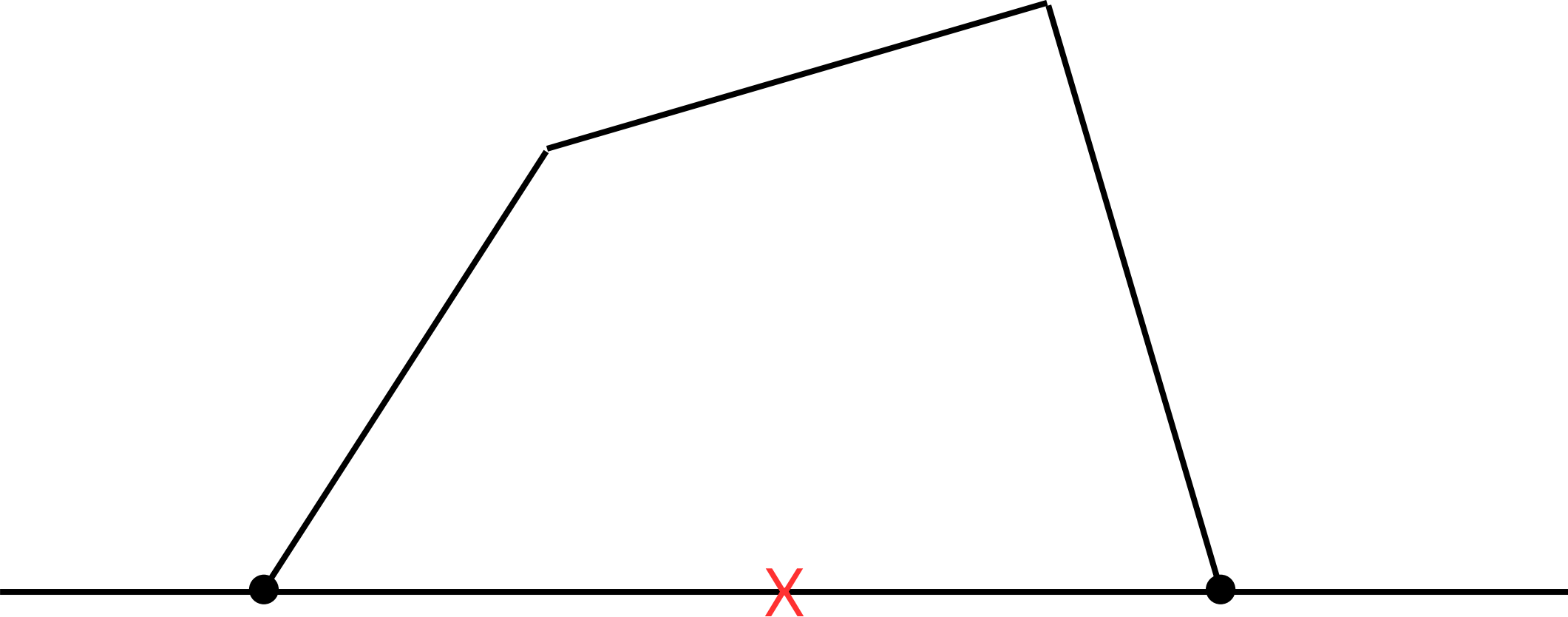
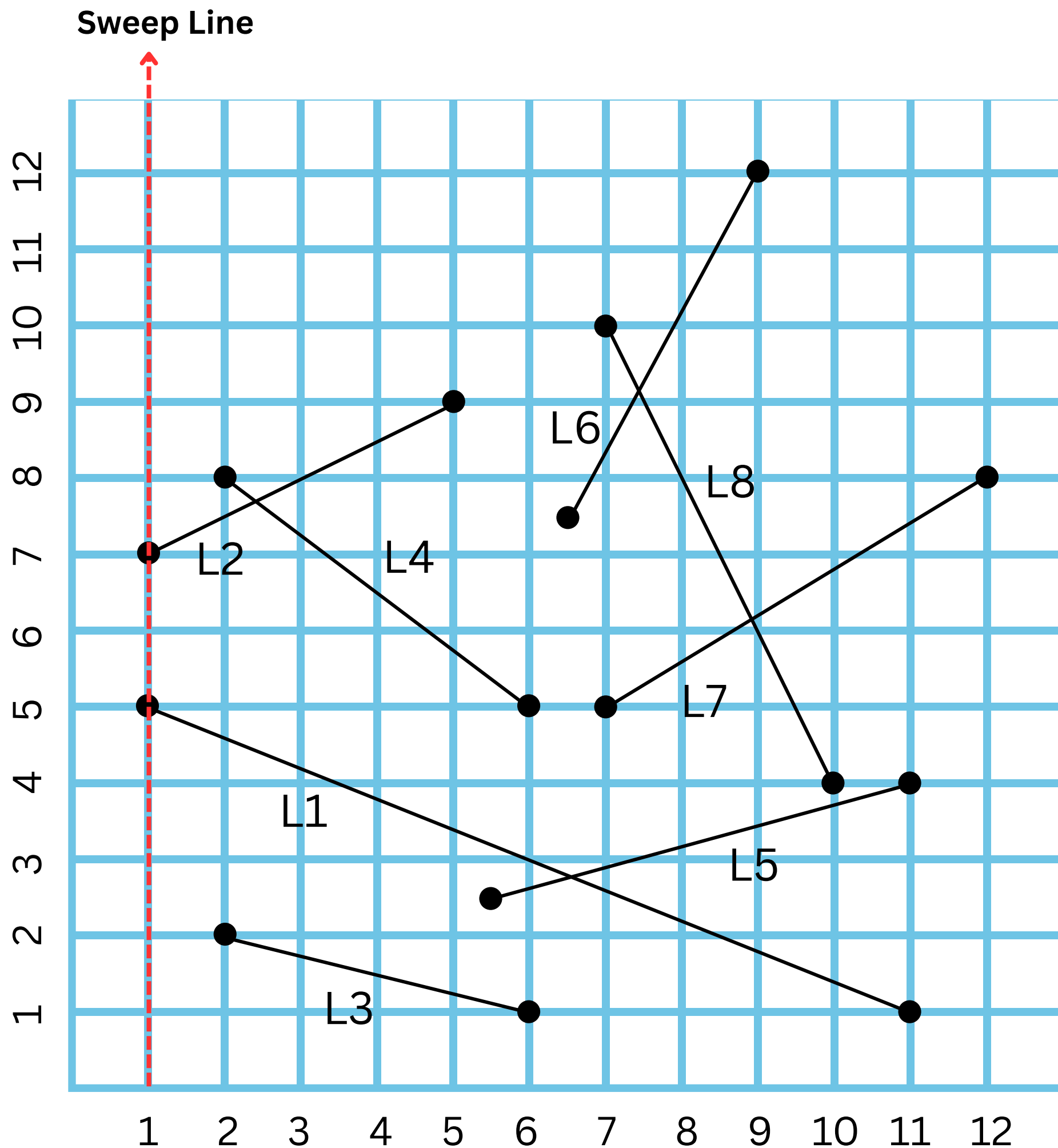




Bentley Ottmann Sweep Line Algorithm

The Data Structures

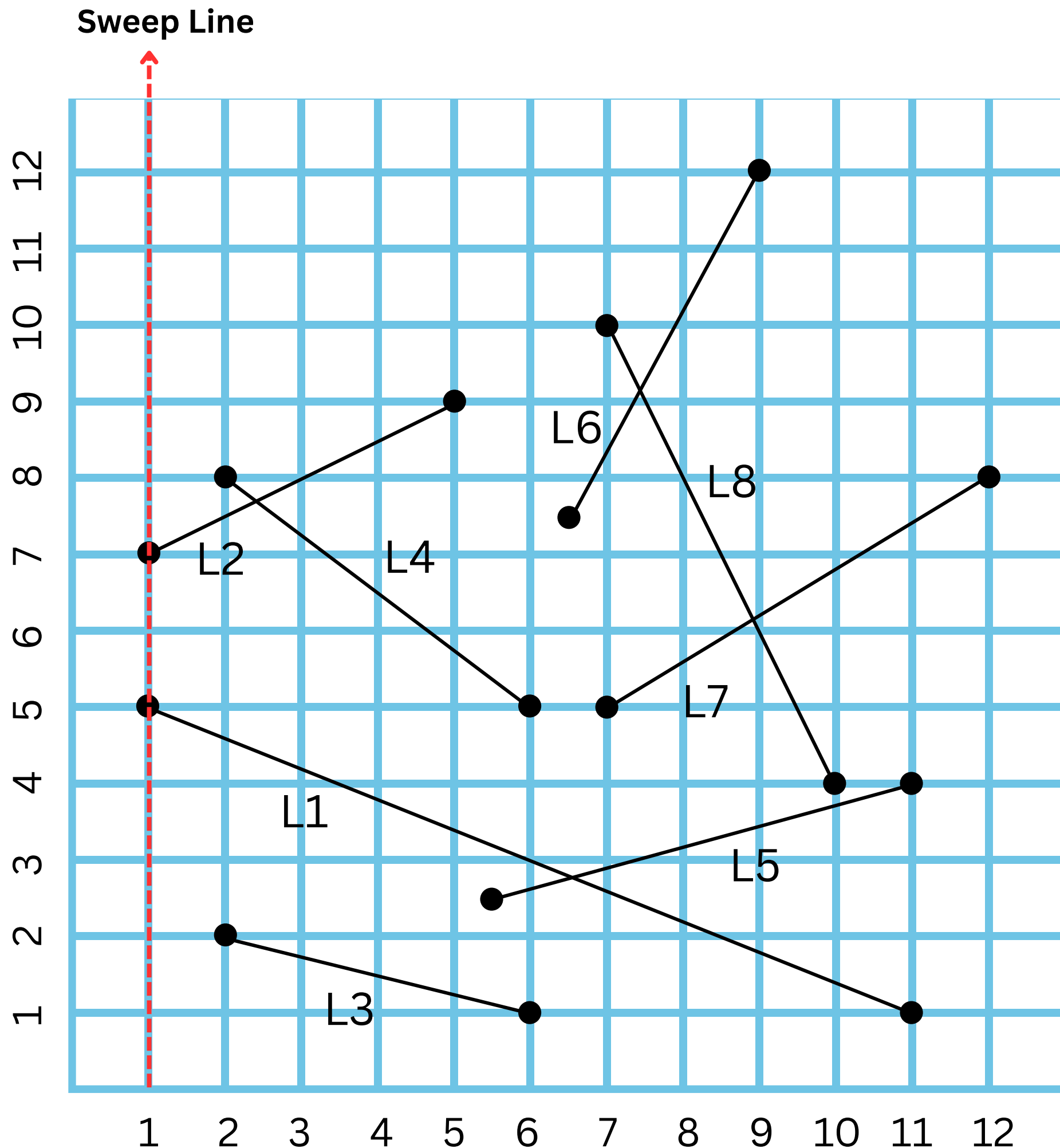


Event Queue

L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
L2_e (5, 9)
L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
L6_b (6.5, 7.5)
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

The Algorithm



Algorithm

If the encountered event is a/an:

A. Left end-point:

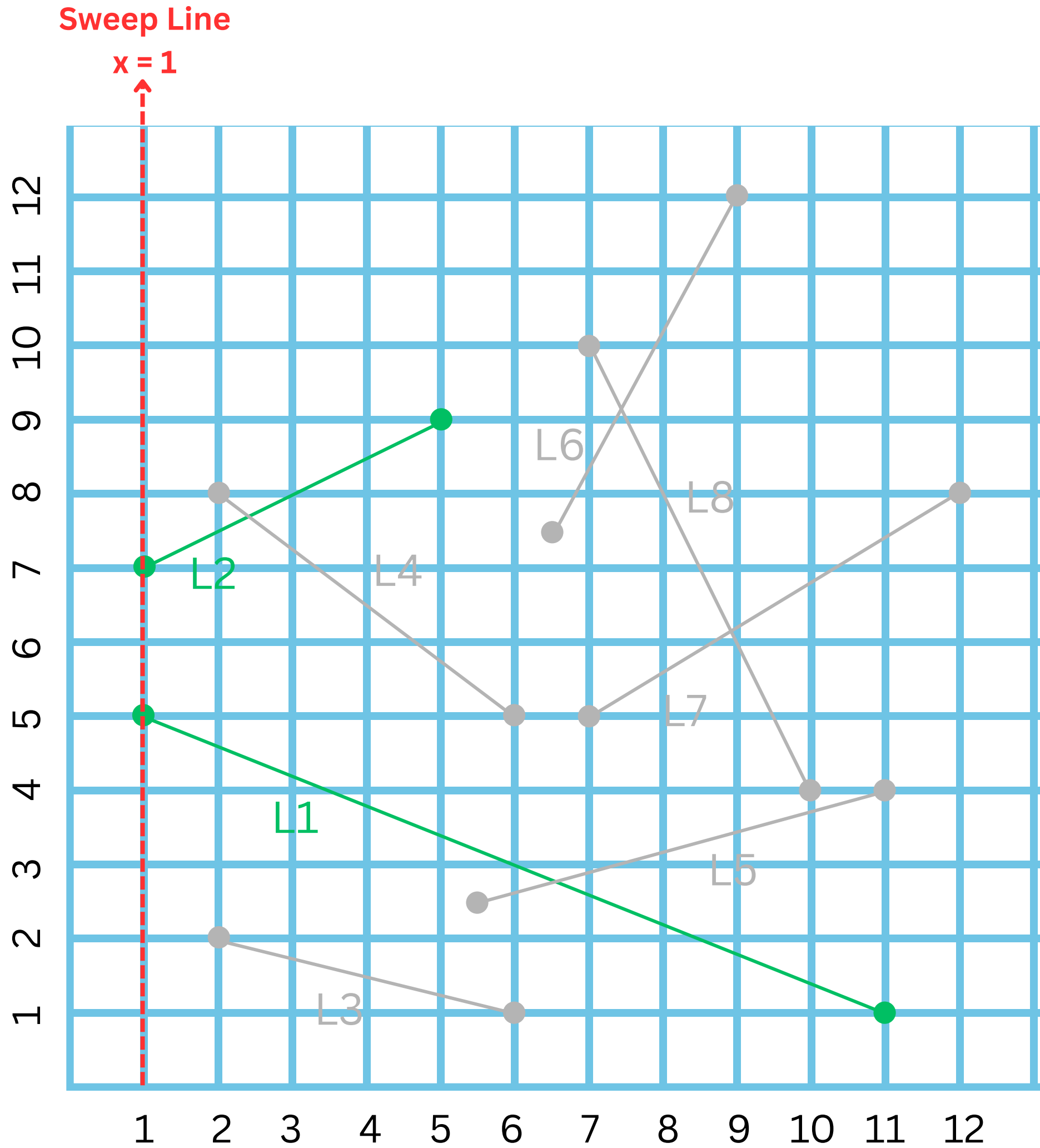
add segment into the status tree and check for intersection with neighbouring segments; if positive insert into the event queue

B. Right end point:

remove segment from the status tree and check intersection among the new neighbours; if positive insert into the event queue

C. Intersection point:

swap the intersecting segments in the status tree and check for intersection among new neighbours



Event Queue

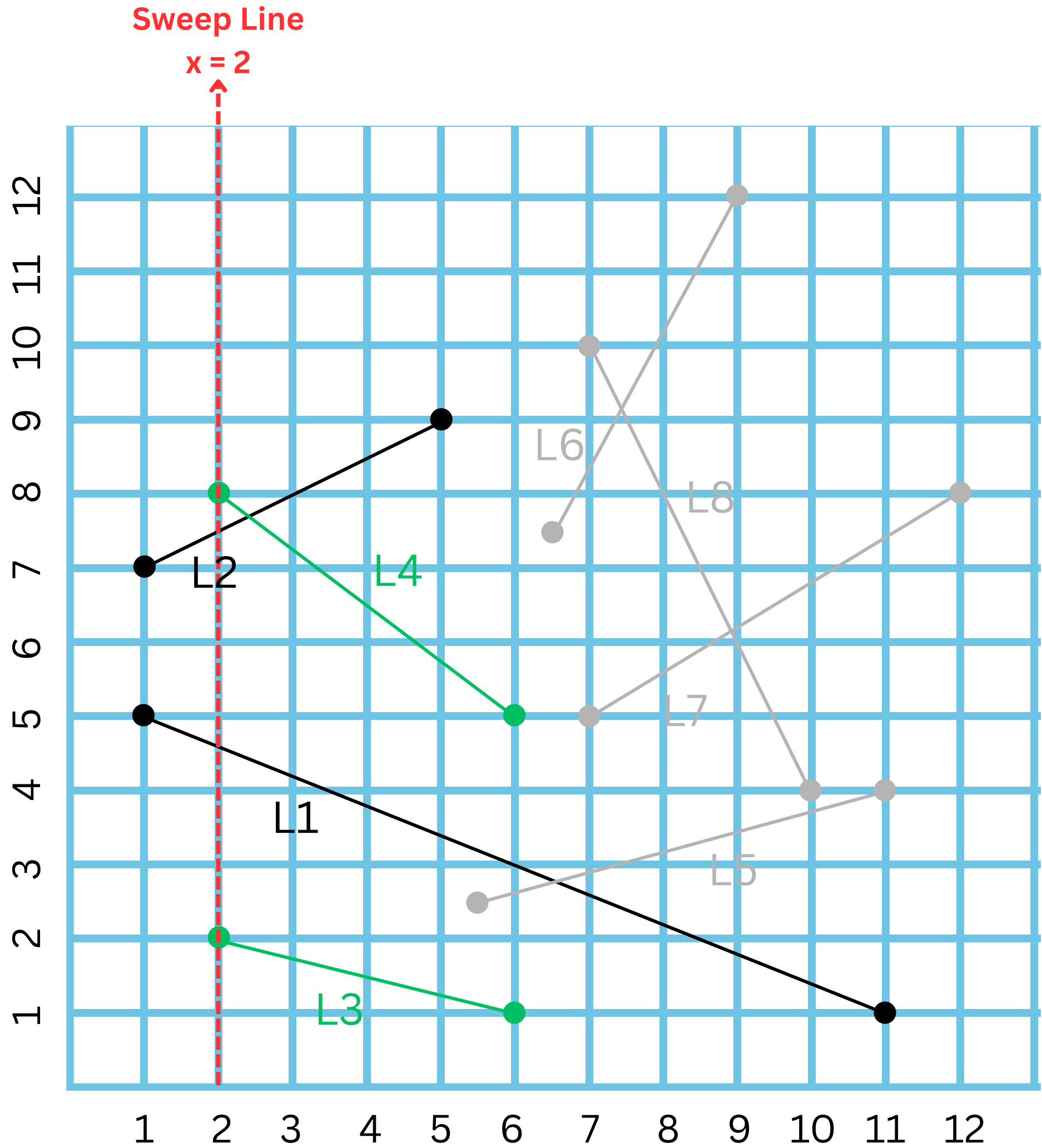
→ **L1b** (1, 5)
L2b (1, 7)
L3b (2, 2)
L4b (2, 8)
L2e (5, 9)
L5b (5.5, 2.5)
L3e (6, 1)
L4e (6, 5)
L6b (6.5, 7.5)
L7b (7, 5)
L8b (7, 10)
L6e (9, 12)
L8e (10, 4)
L1e (11, 1)
L5e (11, 4)
L7e (12, 8)

Status Tree

L2 New
L1 New

Intersection Test

L2 and L1



Event Queue

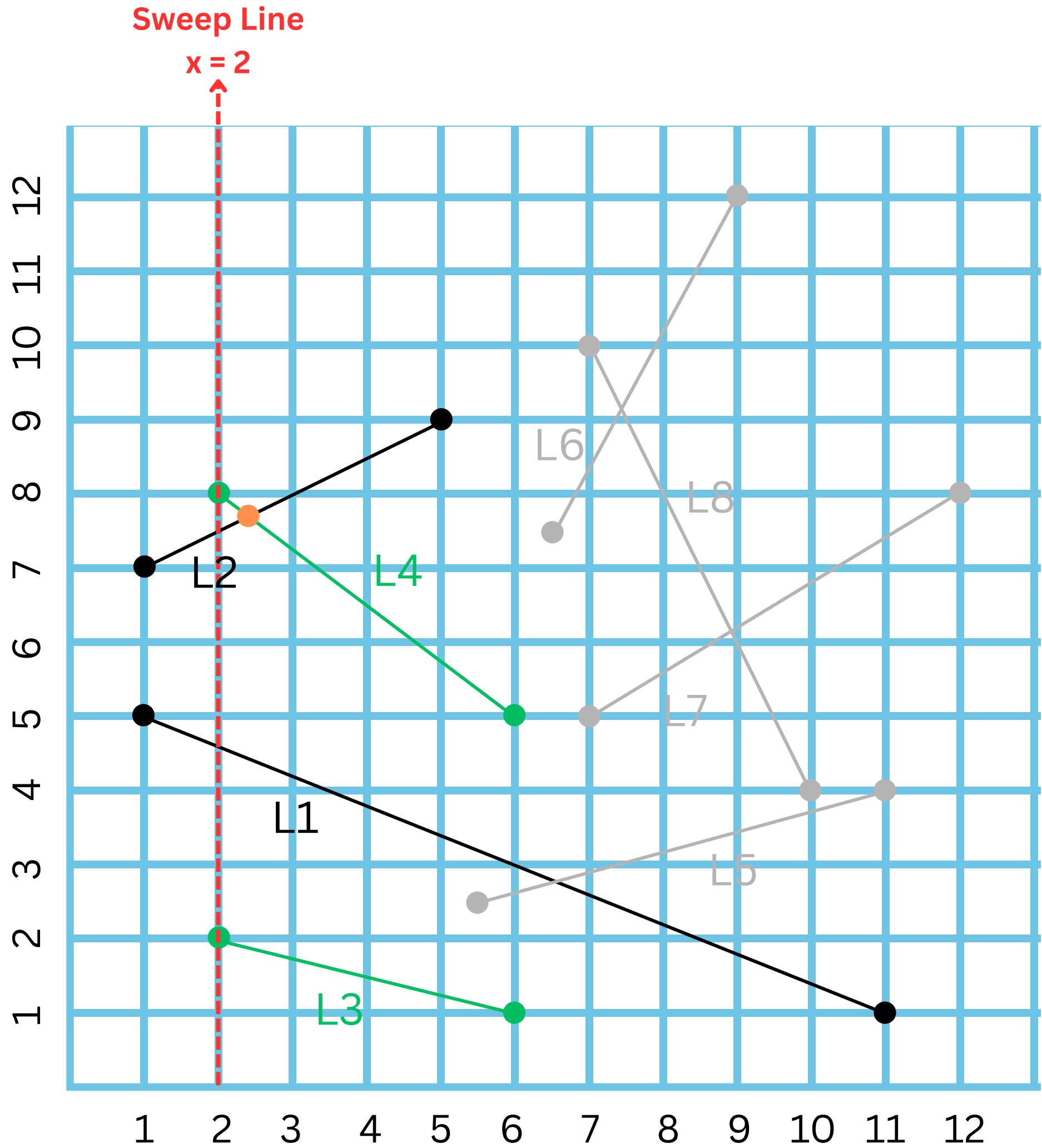
- L1b (1, 5)
- L2b (1, 7)
- L3b (2, 2)**
- L4b (2, 8)**
- L2e (5, 9)
- L5b (5.5, 2.5)
- L3e (6, 1)
- L4e (6, 5)
- L6b (6.5, 7.5)
- L7b (7, 5)
- L8b (7, 10)
- L6e (9, 12)
- L8e (10, 4)
- L1e (11, 1)
- L5e (11, 4)
- L7e (12, 8)

Status Tree

- L4** New
- L2
- L1
- L3** New

Intersection Test

- L4 and L2
- L1 and L3



Event Queue

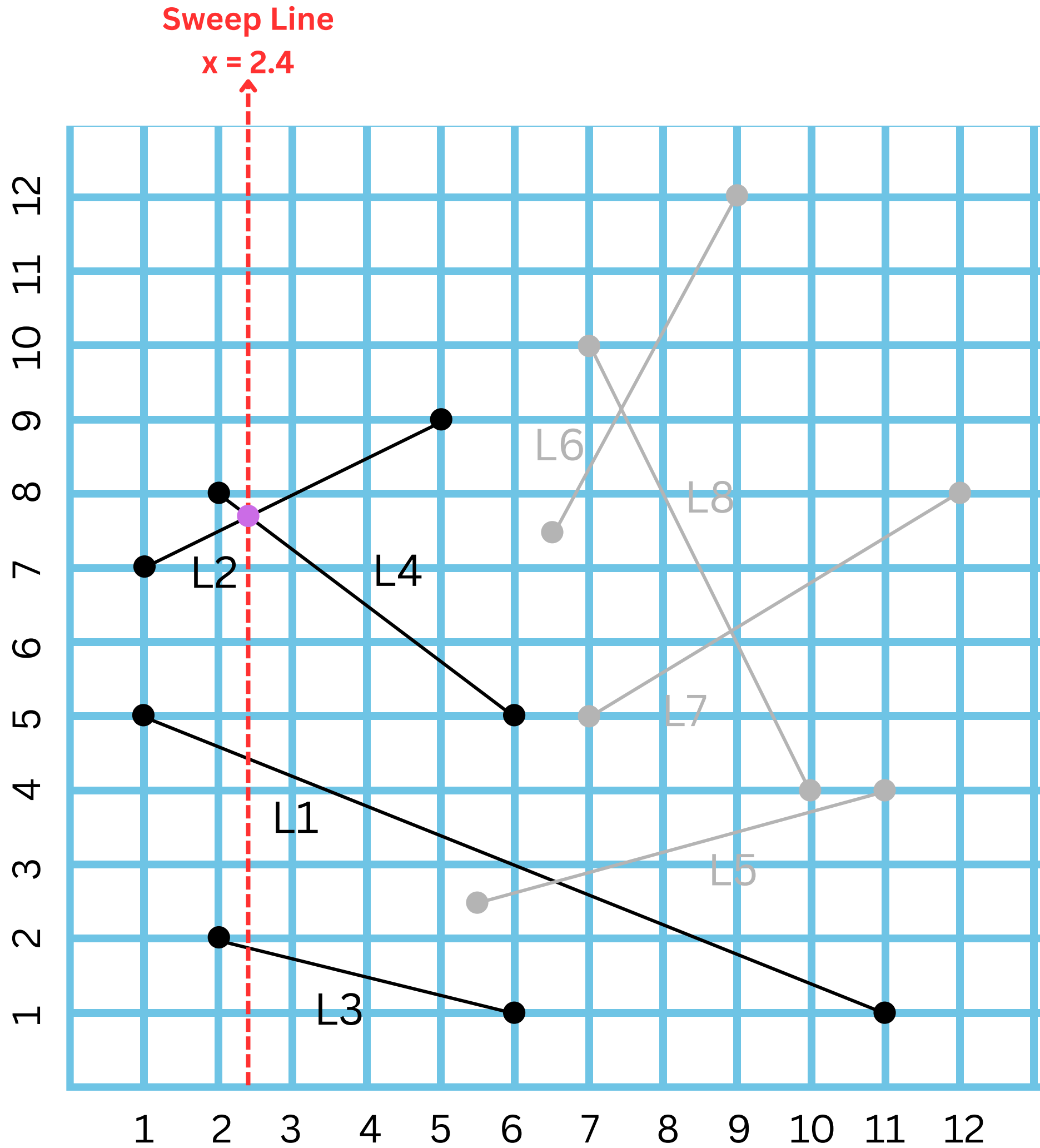
- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- L7_b (7, 5)
- L8_b (7, 10)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L4
- L2
- L1
- L3

Intersection Test

- L4 and L2
- L1 and L3

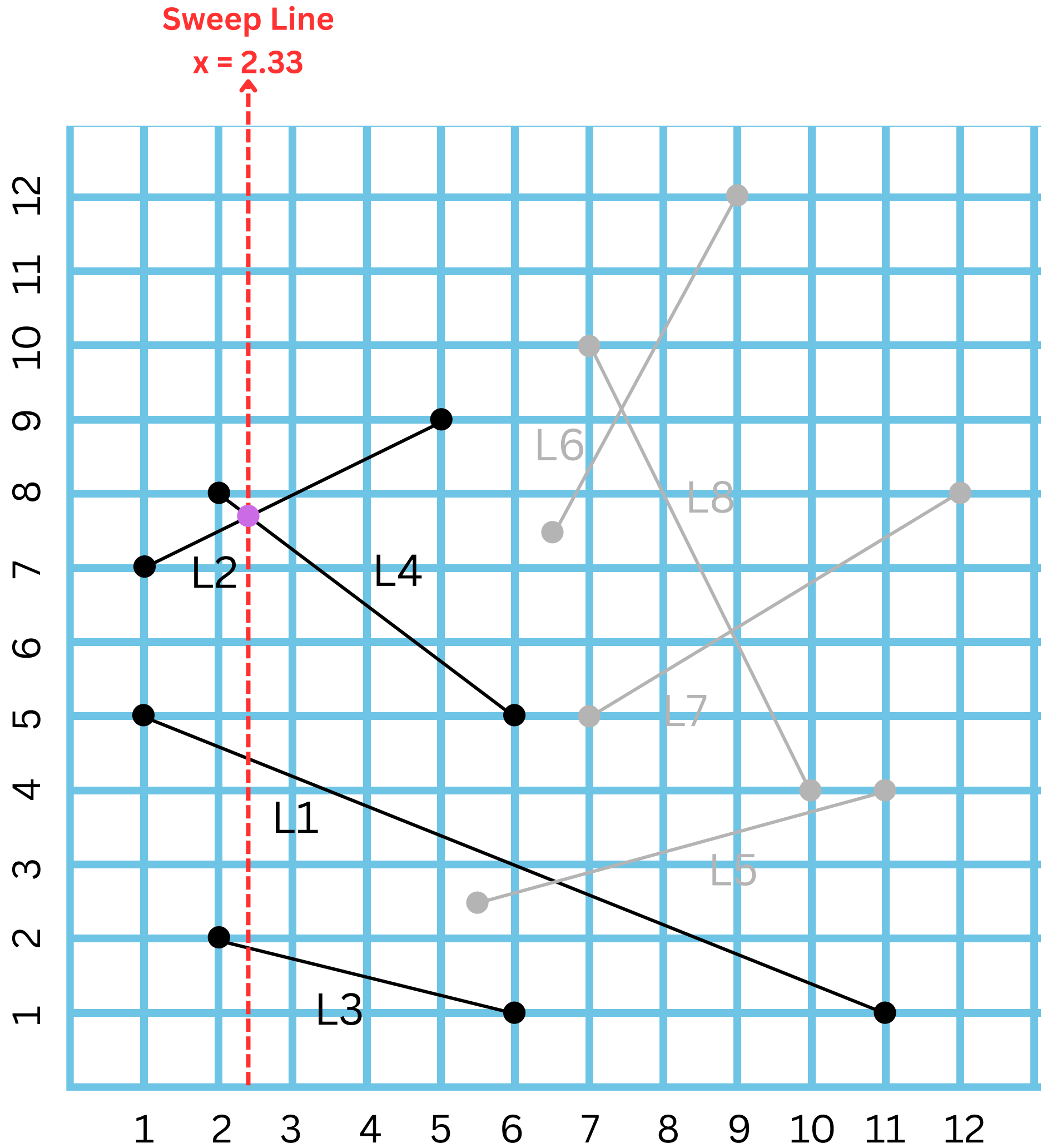


Event Queue

- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- **I[L2, L4] (2.4, 7.7)**
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- L7_b (7, 5)
- L8_b (7, 10)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L4
- L2 Swap
- L1
- L3



Event Queue

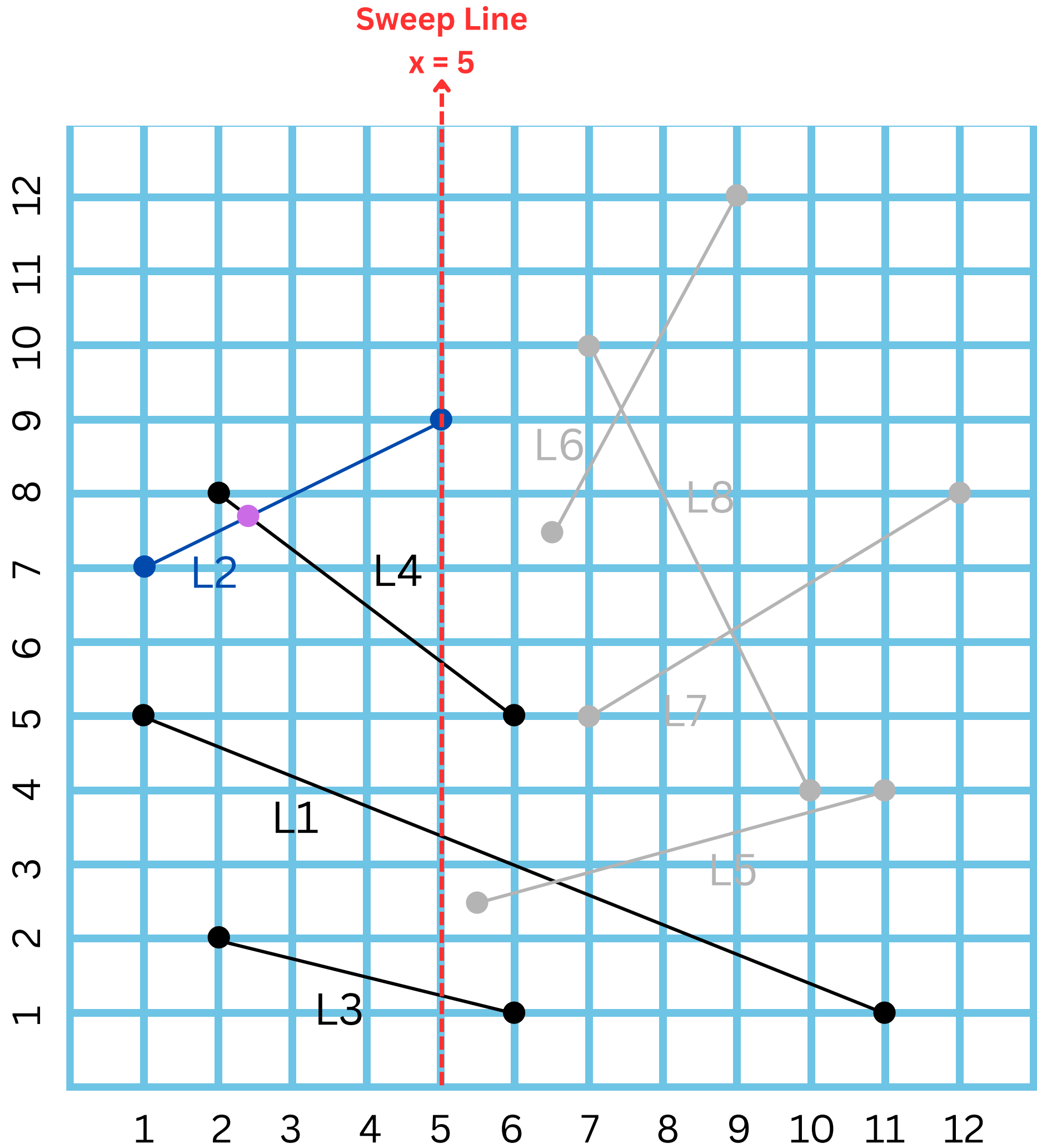
L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
→ **I[L2, L4] (2.4, 7.7)**
L2_e (5, 9)
L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
L6_b (6.5, 7.5)
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

L2
L4
L1
L3

Intersection Test

L4 and L1

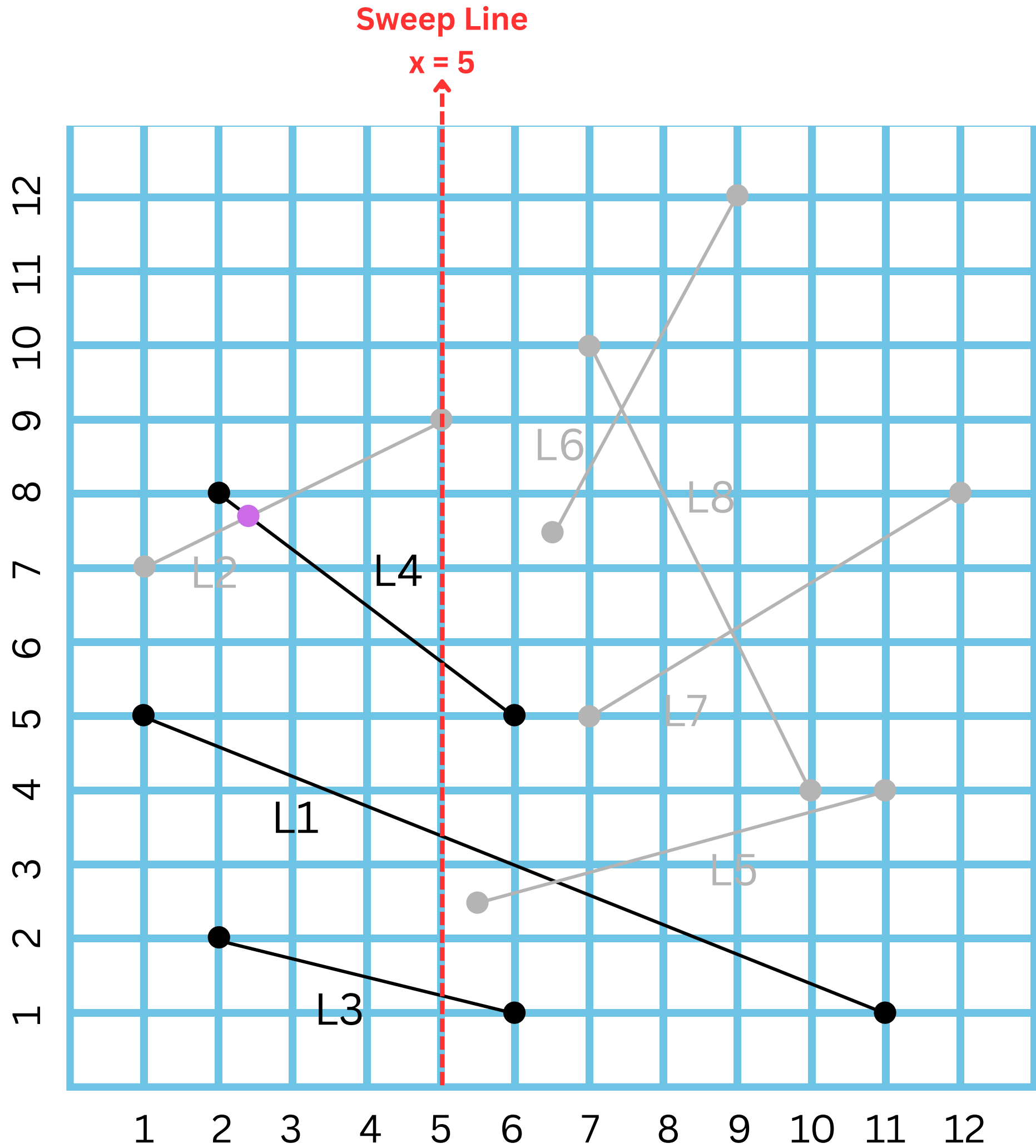


Event Queue

L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
I[L2, L4] (2.4, 7.7)
→ L2_e (5, 9)
L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
L6_b (6.5, 7.5)
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

L2 Delete
L4
L1
L3

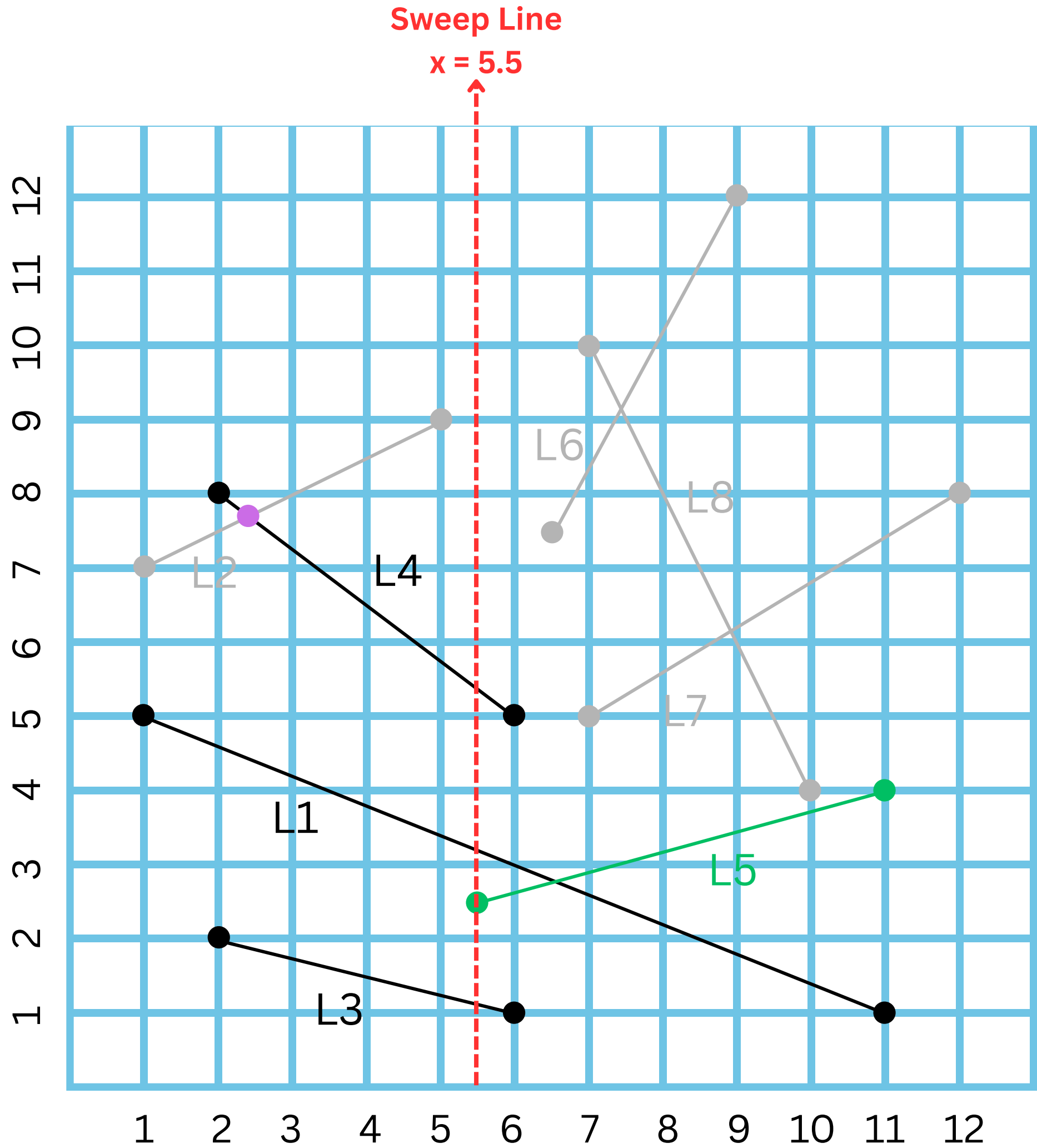


Event Queue

- L1***b* (1, 5)
- L2***b* (1, 7)
- L3***b* (2, 2)
- L4***b* (2, 8)
- I**[**L2**, **L4**] (2.4, 7.7)
- L2***e* (5, 9)
- L5***b* (5.5, 2.5)
- L3***e* (6, 1)
- L4***e* (6, 5)
- L6***b* (6.5, 7.5)
- L7***b* (7, 5)
- L8***b* (7, 10)
- L6***e* (9, 12)
- L8***e* (10, 4)
- L1***e* (11, 1)
- L5***e* (11, 4)
- L7***e* (12, 8)

Status Tree

- L4**
- L1**
- L3**



Event Queue

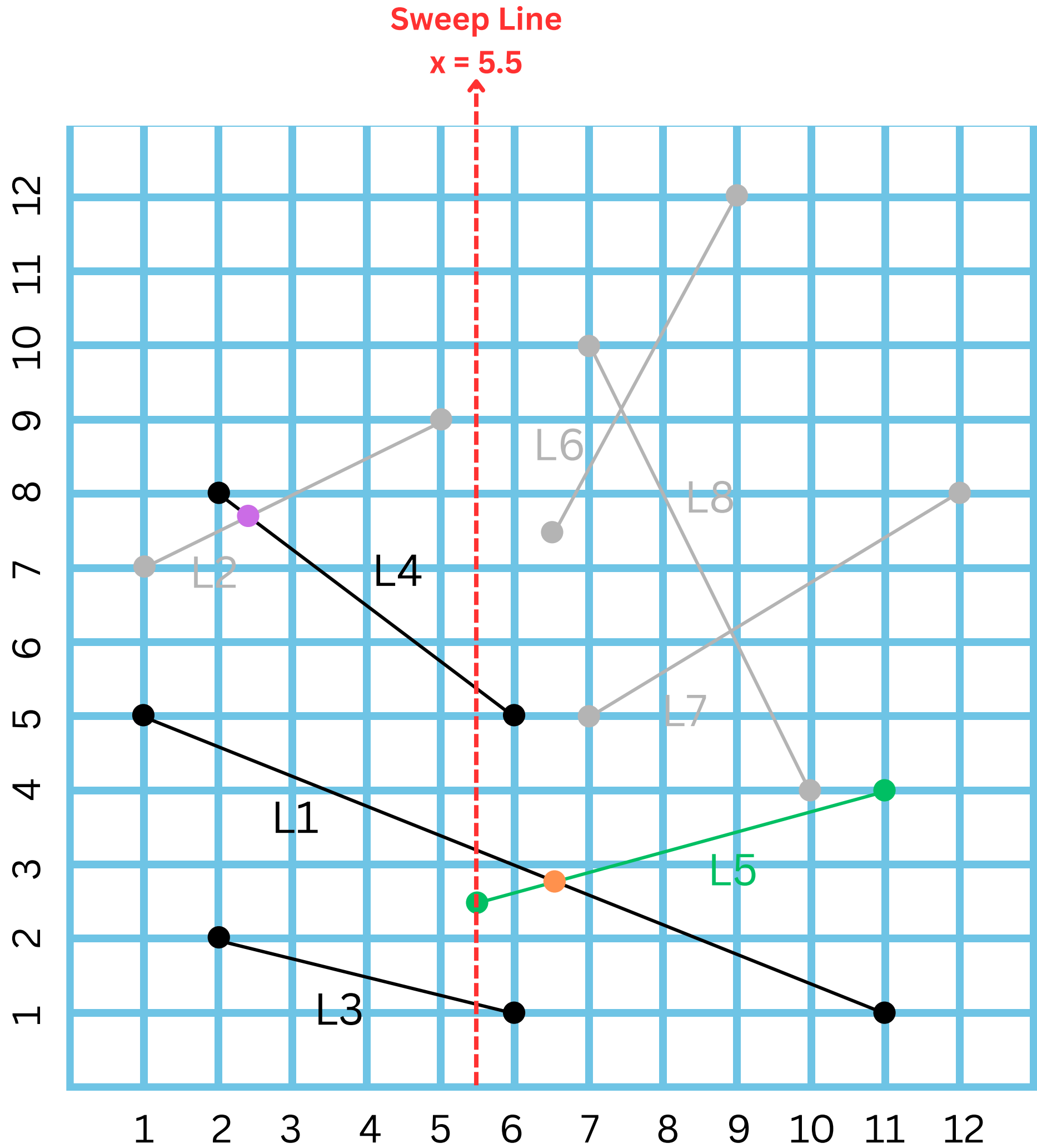
- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- L7_b (7, 5)
- L8_b (7, 10)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L4
- L1
- L5 New
- L3

Intersection Test

- L1 and L5
- L5 and L3



Event Queue

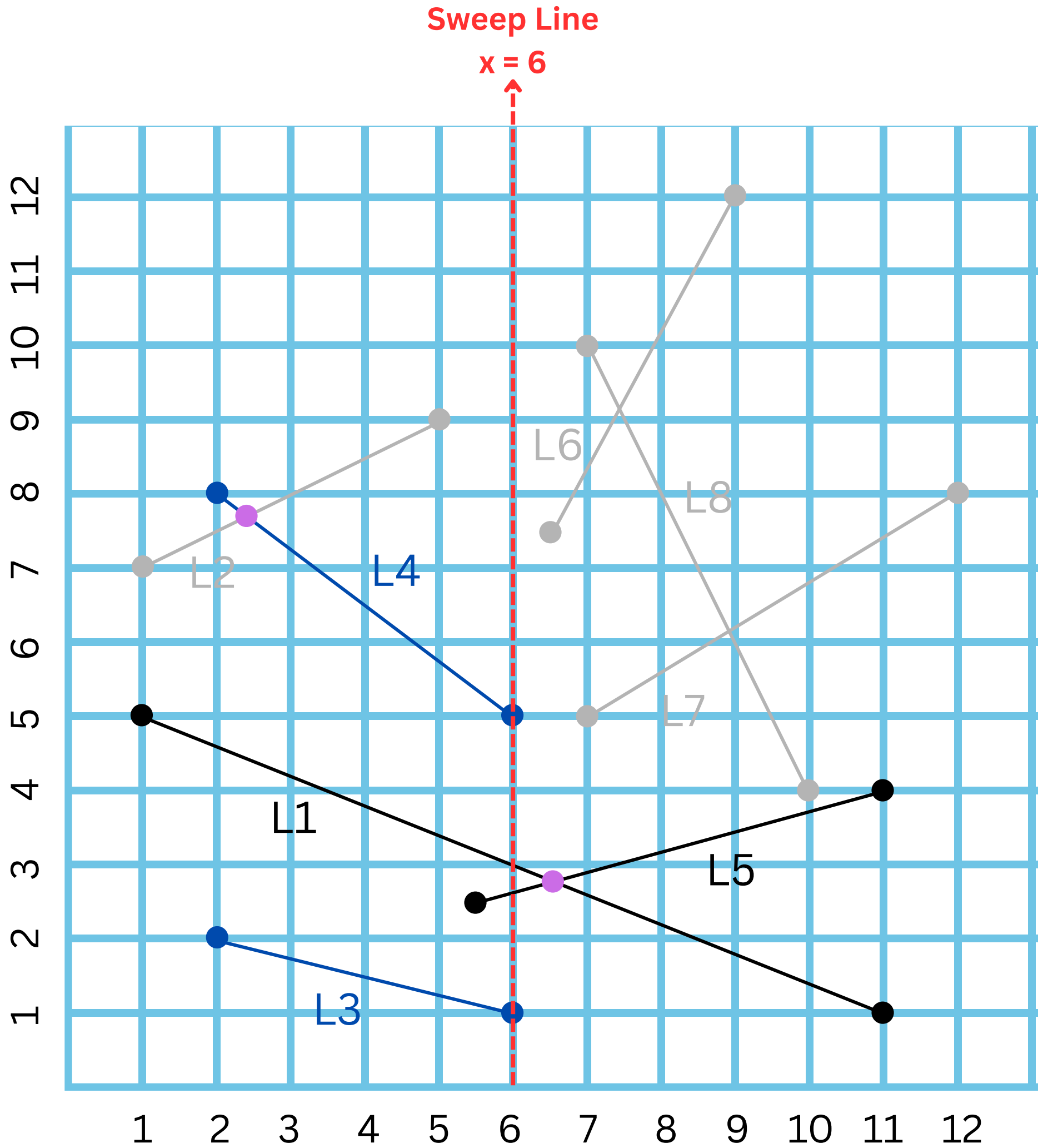
L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
 $I[L2, L4]$ (2.4, 7.7)
L2_e (5, 9)
→ L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
L6_b (6.5, 7.5)
 $I[L1, L5]$ (6.54, 2.78)
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

L4
L1
L5
L3

Intersection Test

L1 and L5
L5 and L3

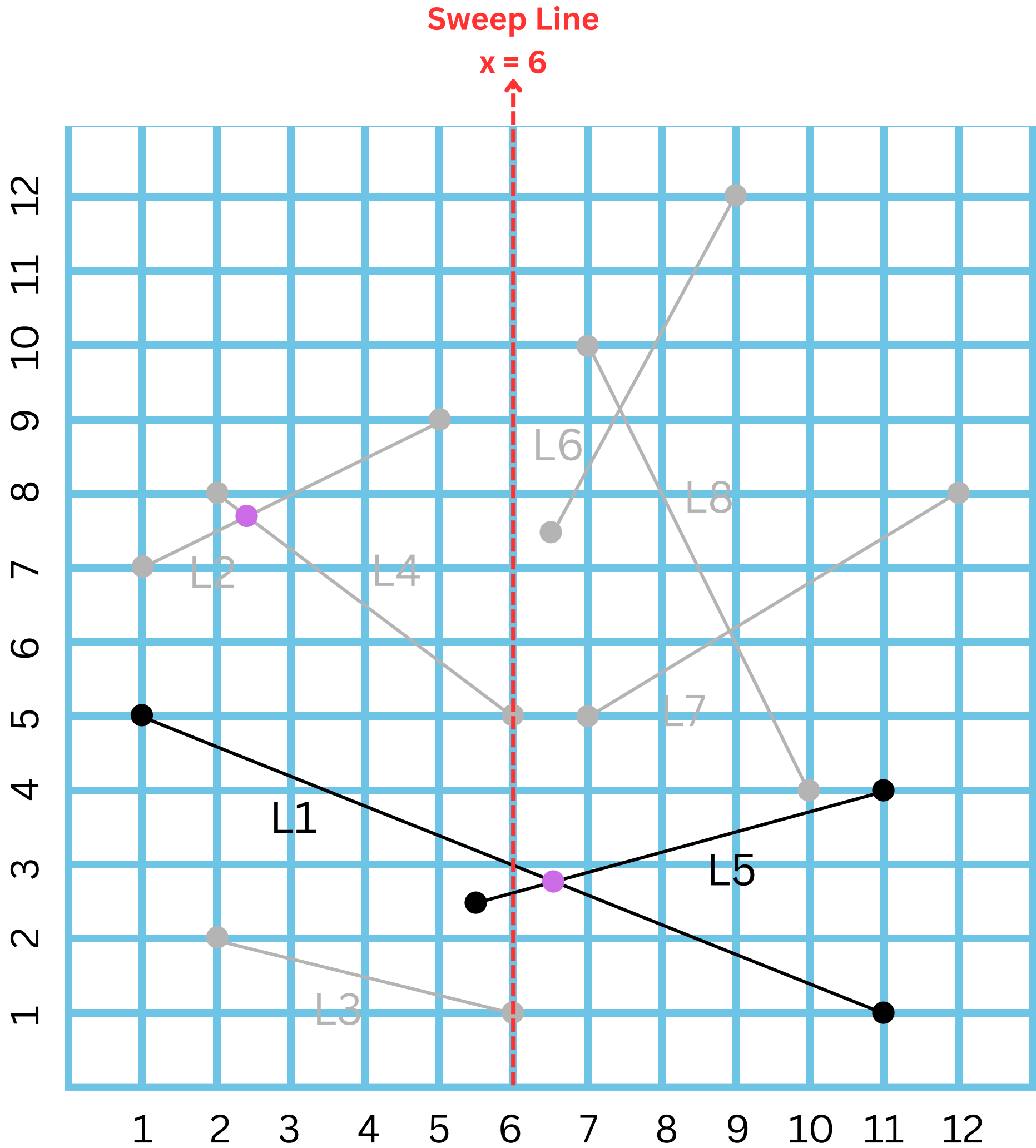


Event Queue

- L1b (1, 5)
- L2b (1, 7)
- L3b (2, 2)
- L4b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2e (5, 9)
- L5b (5.5, 2.5)
- L3e (6, 1)
- L4e (6, 5)
- L6b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7b (7, 5)
- L8b (7, 10)
- L6e (9, 12)
- L8e (10, 4)
- L1e (11, 1)
- L5e (11, 4)
- L7e (12, 8)

Status Tree

- L4 Delete
- L1
- L5
- L3 Delete

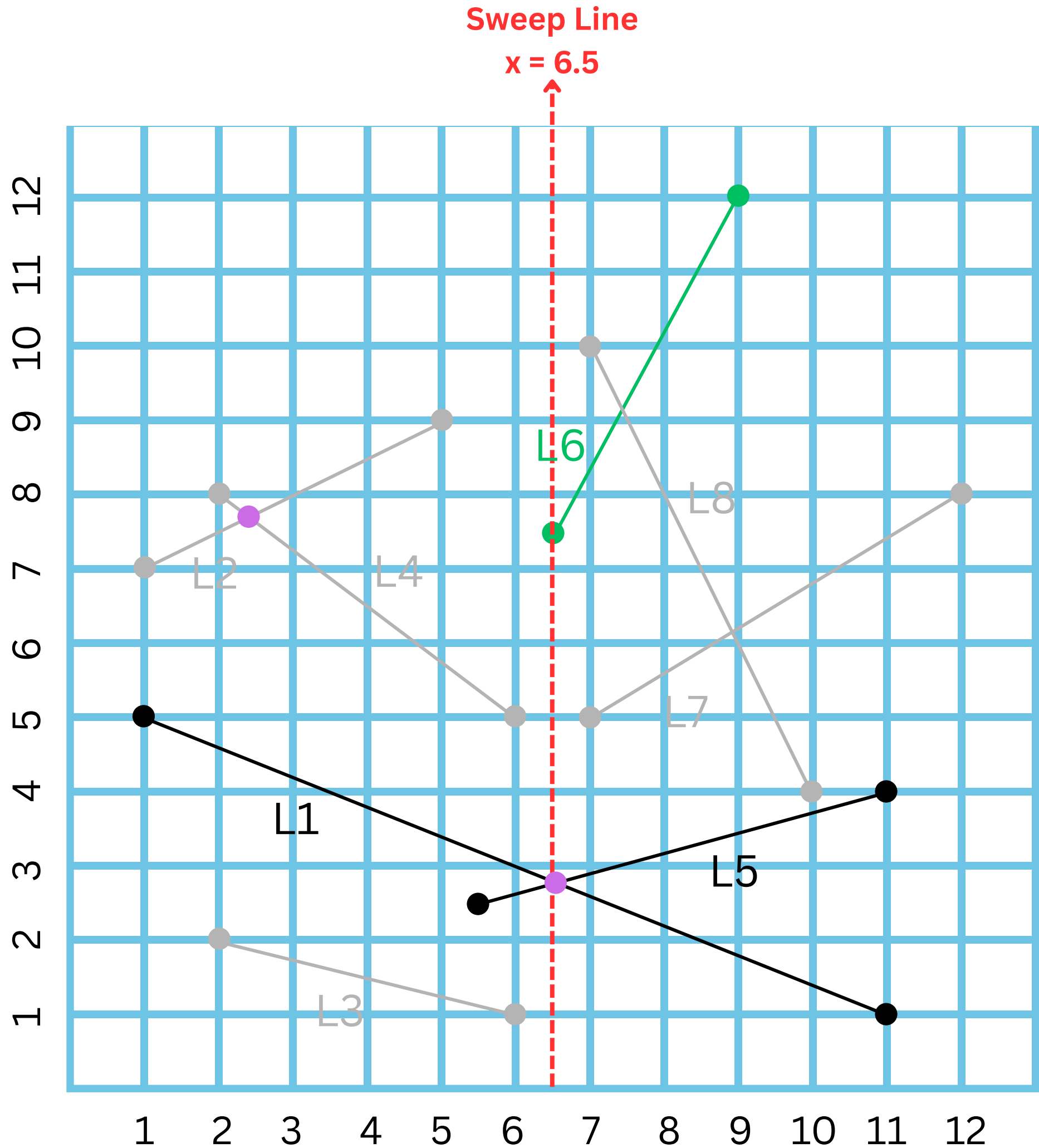


Event Queue

L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
 $I[L2, L4] (2.4, 7.7)$
L2_e (5, 9)
L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
L6_b (6.5, 7.5)
 $I[L1, L5] (6.54, 2.78)$
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

L1
L5



Event Queue

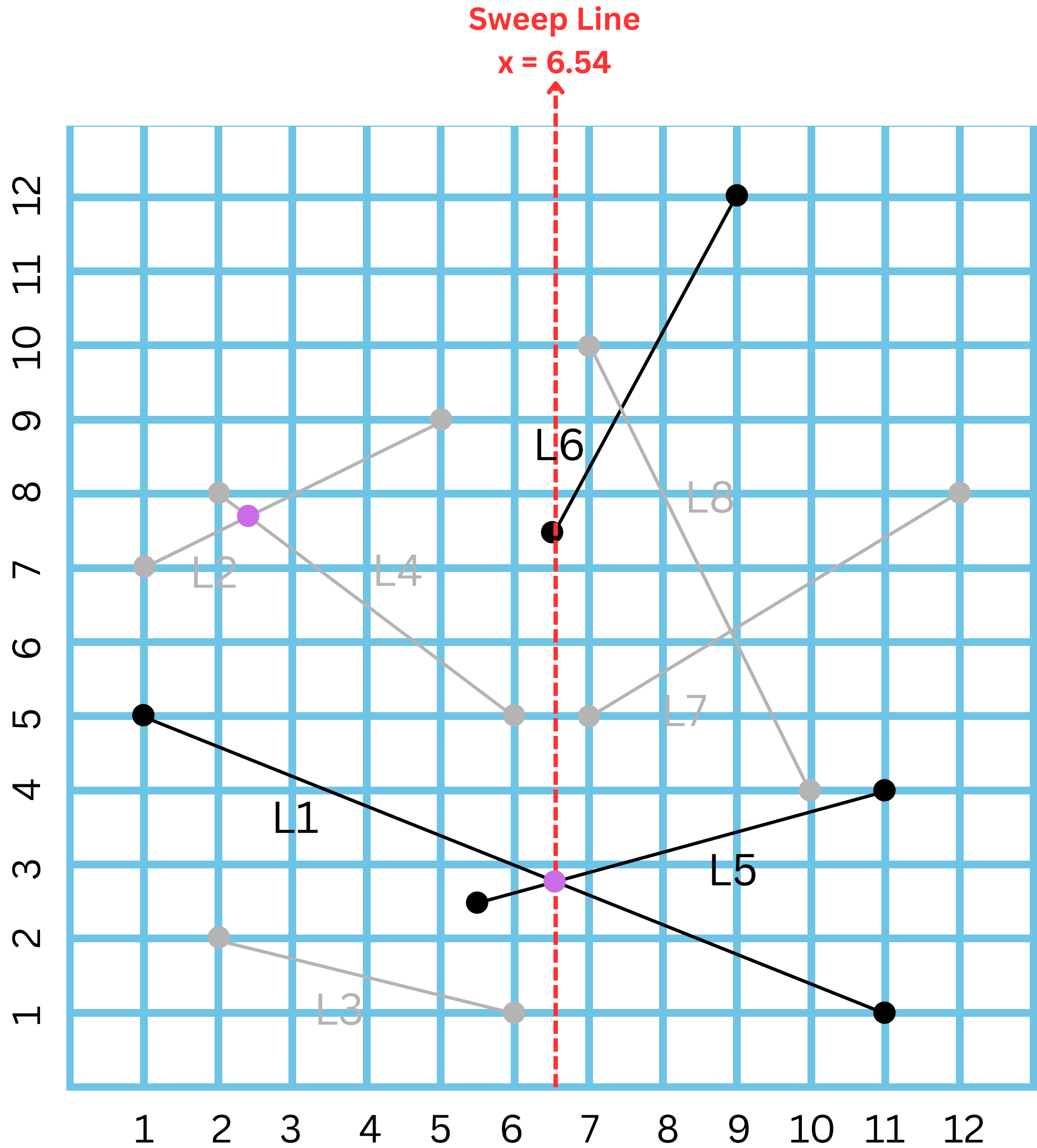
L1_b (1, 5)
L2_b (1, 7)
L3_b (2, 2)
L4_b (2, 8)
I[L2, L4] (2.4, 7.7)
L2_e (5, 9)
L5_b (5.5, 2.5)
L3_e (6, 1)
L4_e (6, 5)
→ L6_b (6.5, 7.5)
I[L1, L5] (6.54, 2.78)
L7_b (7, 5)
L8_b (7, 10)
L6_e (9, 12)
L8_e (10, 4)
L1_e (11, 1)
L5_e (11, 4)
L7_e (12, 8)

Status Tree

L6 New
L1
L5

Intersection Test

L6 and L1

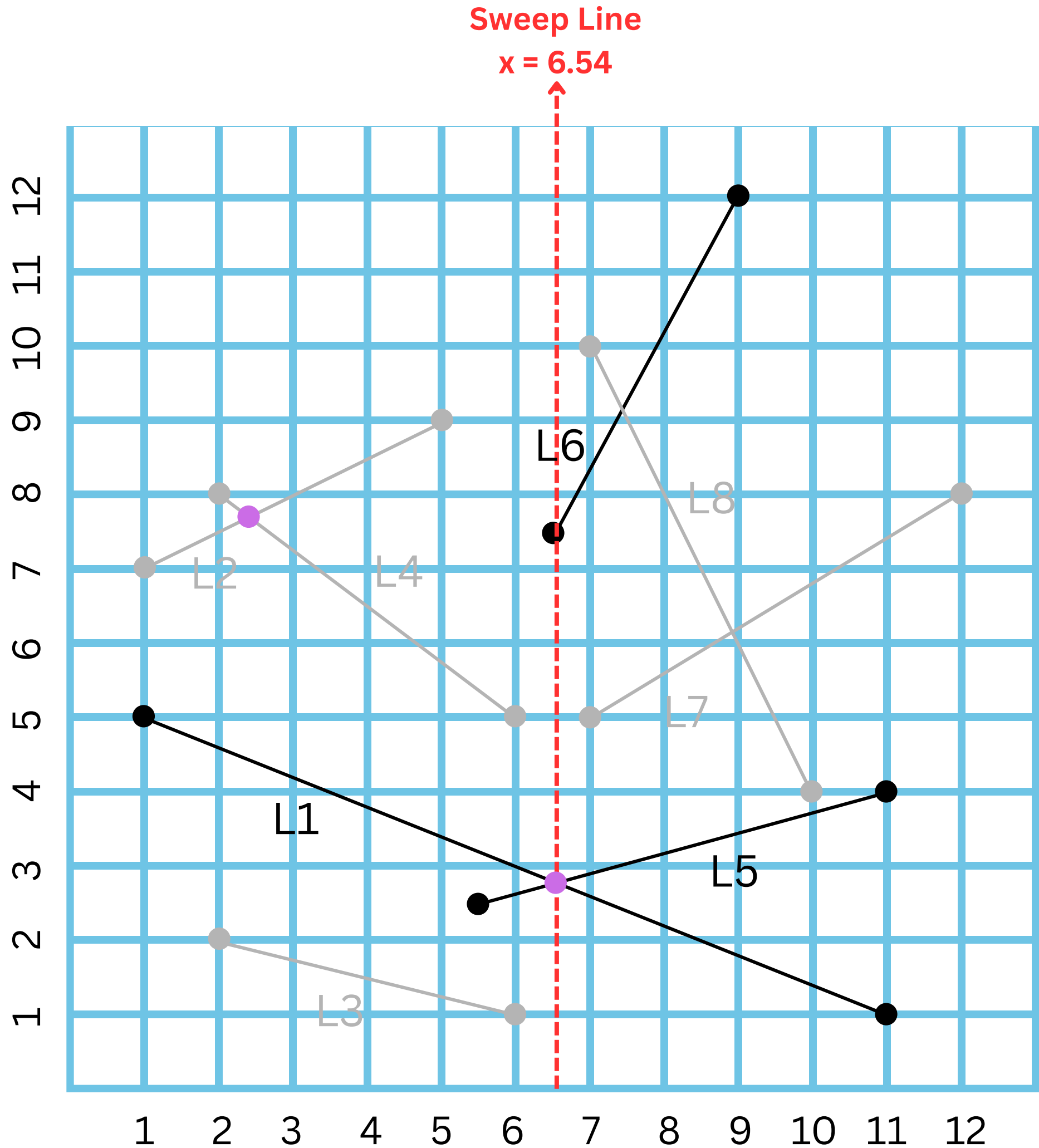


Event Queue

L1*b* (1, 5)
L2*b* (1, 7)
L3*b* (2, 2)
L4*b* (2, 8)
I[**L2**, **L4**] (2.4, 7.7)
L2*e* (5, 9)
L5*b* (5.5, 2.5)
L3*e* (6, 1)
L4*e* (6, 5)
L6*b* (6.5, 7.5)
→ I[**L1**, **L5**] (6.54, 2.78)
L7*b* (7, 5)
L8*b* (7, 10)
L6*e* (9, 12)
L8*e* (10, 4)
L1*e* (11, 1)
L5*e* (11, 4)
L7*e* (12, 8)

Status Tree

L6
L1
L5 Swap



Event Queue

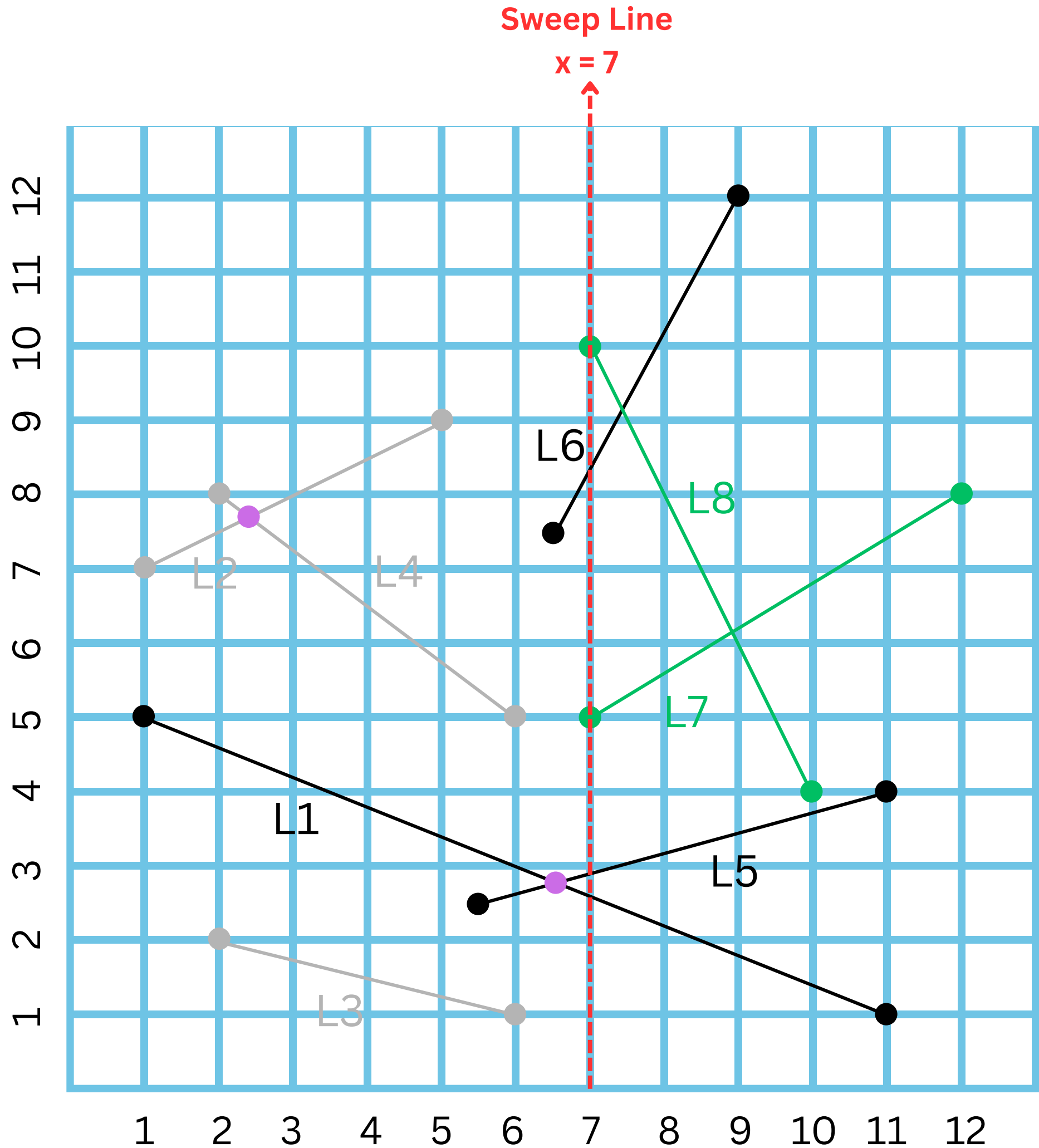
- L1b (1, 5)
- L2b (1, 7)
- L3b (2, 2)
- L4b (2, 8)
- $I[L2, L4]$ (2.4, 7.7)
- L2e (5, 9)
- L5b (5.5, 2.5)
- L3e (6, 1)
- L4e (6, 5)
- L6b (6.5, 7.5)
- $\rightarrow I[L1, L5]$ (6.54, 2.78)
- L7b (7, 5)
- L8b (7, 10)
- L6e (9, 12)
- L8e (10, 4)
- L1e (11, 1)
- L5e (11, 4)
- L7e (12, 8)

Status Tree

- L6
- L5
- L1

Intersection Test

L6 and L5



Event Queue

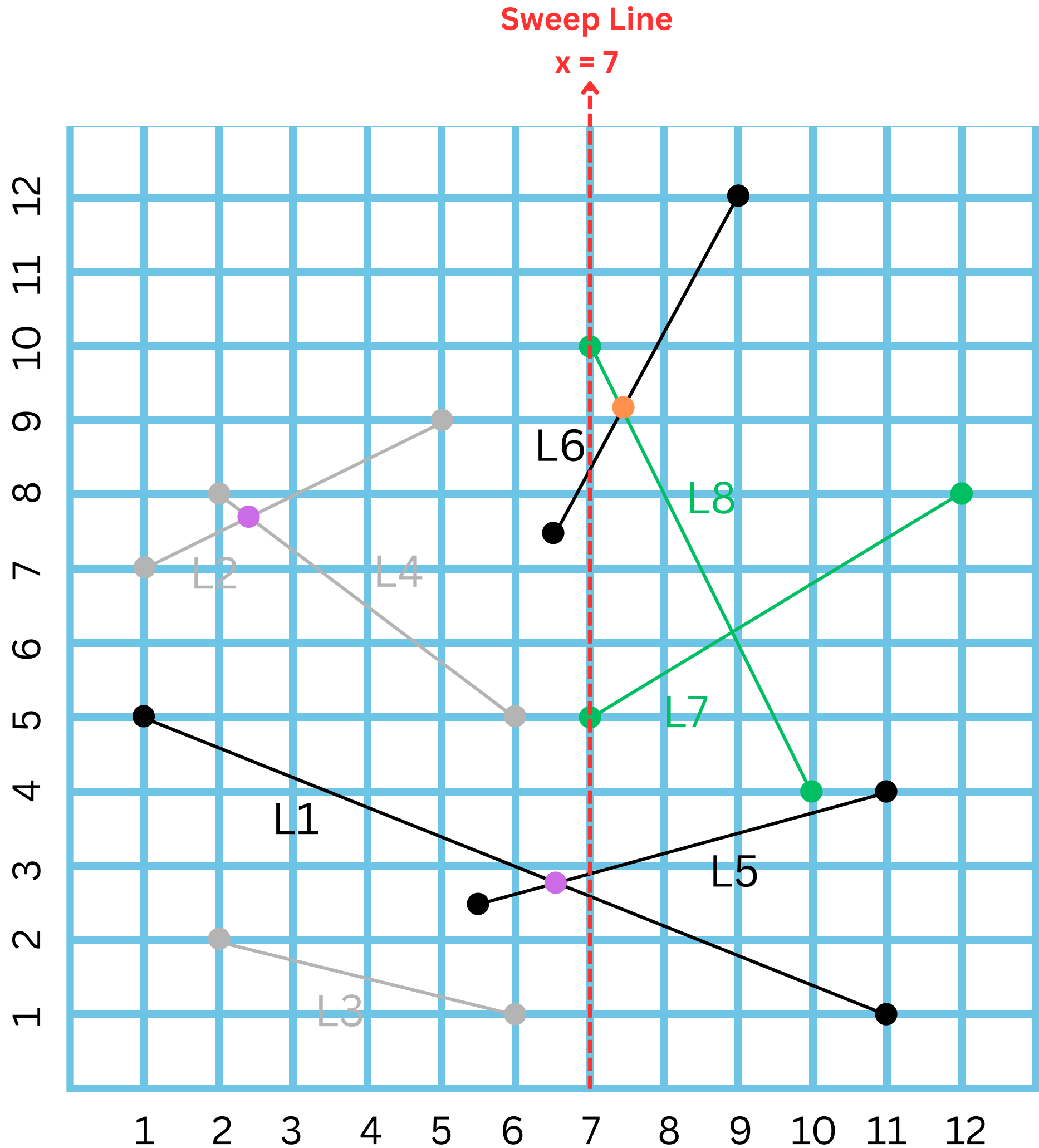
L1b (1, 5)
L2b (1, 7)
L3b (2, 2)
L4b (2, 8)
I[L2, L4] (2.4, 7.7)
L2e (5, 9)
L5b (5.5, 2.5)
L3e (6, 1)
L4e (6, 5)
L6b (6.5, 7.5)
I[L1, L5] (6.54, 2.78)
→ L7b (7, 5)
L8b (7, 10)
L6e (9, 12)
L8e (10, 4)
L1e (11, 1)
L5e (11, 4)
L7e (12, 8)

Status Tree

L8 New
L6
L7 New
L5
L1

Intersection Test

L8 and L6
L6 and L7
L7 and L5



Event Queue

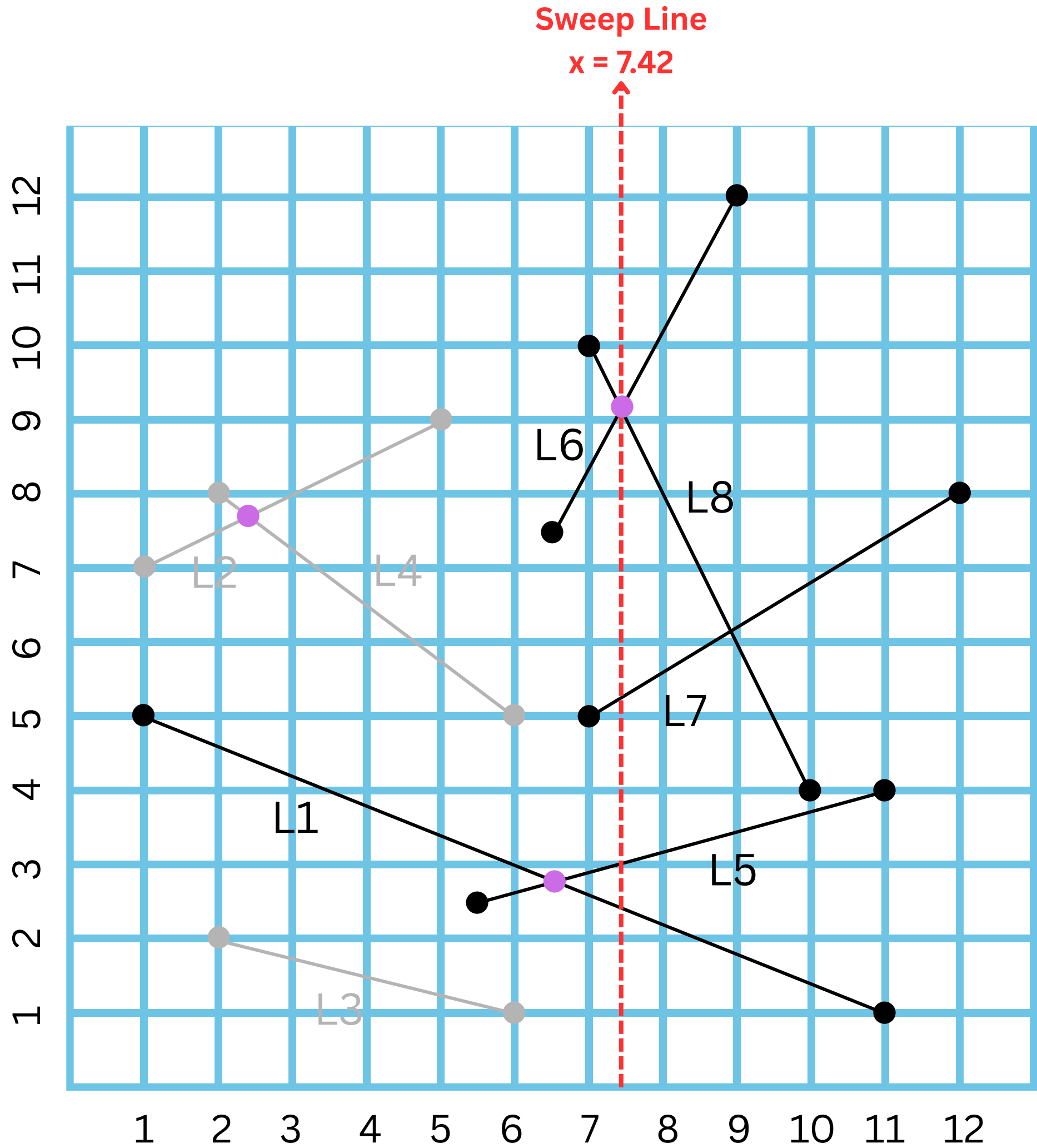
L1*b* (1, 5)
L2*b* (1, 7)
L3*b* (2, 2)
L4*b* (2, 8)
I[**L2**, **L4**] (2.4, 7.7)
L2*e* (5, 9)
L5*b* (5.5, 2.5)
L3*e* (6, 1)
L4*e* (6, 5)
L6*b* (6.5, 7.5)
I[**L1**, **L5**] (6.54, 2.78)
L7*b* (7, 5)
L8*b* (7, 10)
I[**L8**, **L6**] (7.42, 9.16)
L6*e* (9, 12)
L8*e* (10, 4)
L1*e* (11, 1)
L5*e* (11, 4)
L7*e* (12, 8)

Status Tree

L8
L6
L7
L5
L1

Intersection Test

L8 and L6
L6 and L7
L7 and L5

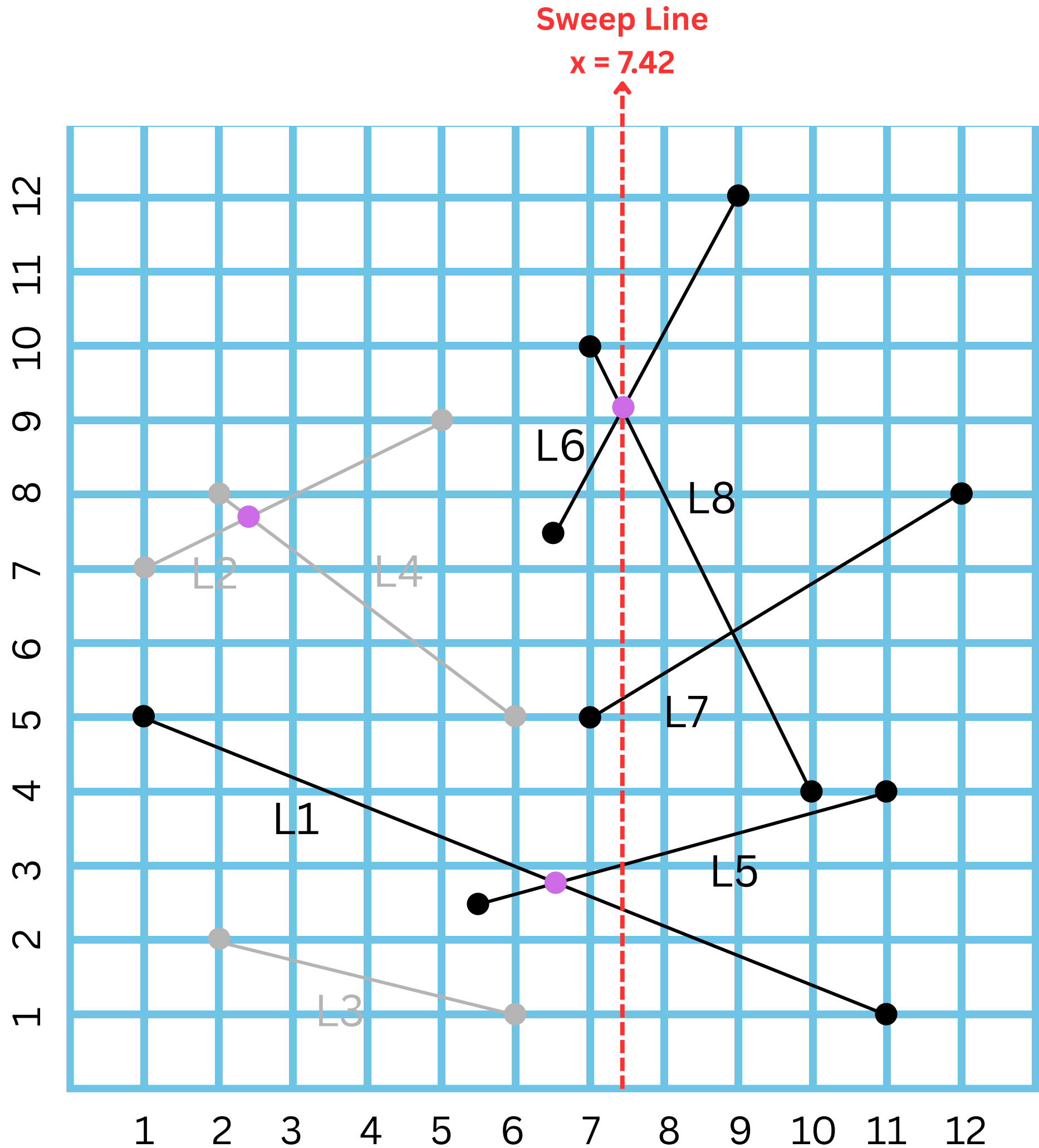


Event Queue

L1*b* (1, 5)
L2*b* (1, 7)
L3*b* (2, 2)
L4*b* (2, 8)
I[**L2**, **L4**] (2.4, 7.7)
L2*e* (5, 9)
L5*b* (5.5, 2.5)
L3*e* (6, 1)
L4*e* (6, 5)
L6*b* (6.5, 7.5)
I[**L1**, **L5**] (6.54, 2.78)
L7*b* (7, 5)
L8*b* (7, 10)
→ **I**[**L8**, **L6**] (7.42, 9.16)
L6*e* (9, 12)
L8*e* (10, 4)
L1*e* (11, 1)
L5*e* (11, 4)
L7*e* (12, 8)

Status Tree

L8
L6 Swap
L7
L5
L1



Event Queue

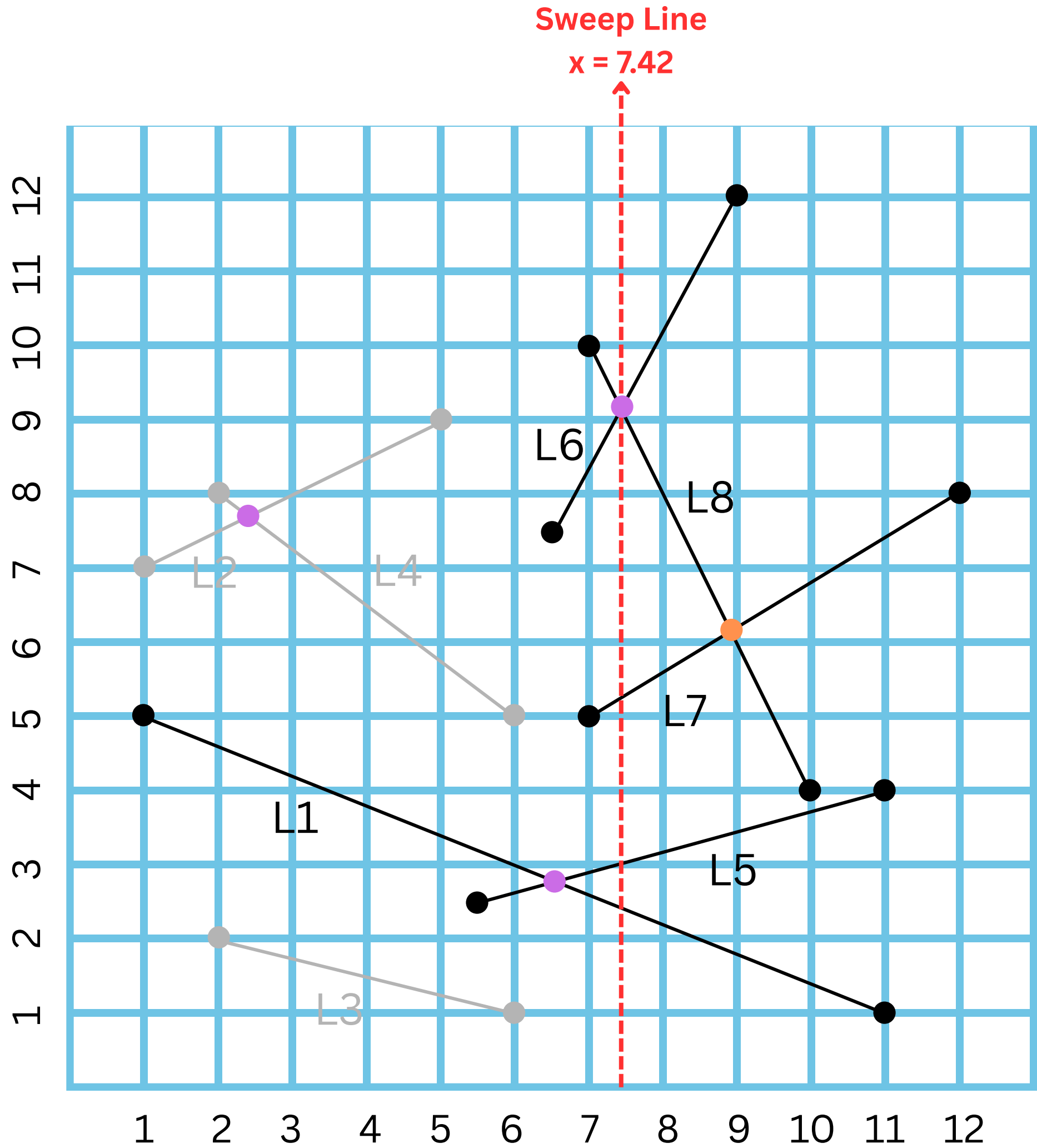
- L1b (1, 5)
- L2b (1, 7)
- L3b (2, 2)
- L4b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2e (5, 9)
- L5b (5.5, 2.5)
- L3e (6, 1)
- L4e (6, 5)
- L6b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7b (7, 5)
- L8b (7, 10)
- I[L8, L6] (7.42, 9.16)
- L6e (9, 12)
- L8e (10, 4)
- L1e (11, 1)
- L5e (11, 4)
- L7e (12, 8)

Status Tree

- L6
- L8
- L7
- L5
- L1

Intersection Test

L8 and L7



Event Queue

- L1b (1, 5)
- L2b (1, 7)
- L3b (2, 2)
- L4b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2e (5, 9)
- L5b (5.5, 2.5)
- L3e (6, 1)
- L4e (6, 5)
- L6b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7b (7, 5)
- L8b (7, 10)
- I[L8, L6] (7.42, 9.16)
- I[L8, L7] (8.92, 6.15)
- L6e (9, 12)
- L8e (10, 4)
- L1e (11, 1)
- L5e (11, 4)
- L7e (12, 8)

Status Tree

- L6
- L8
- L7
- L5
- L1

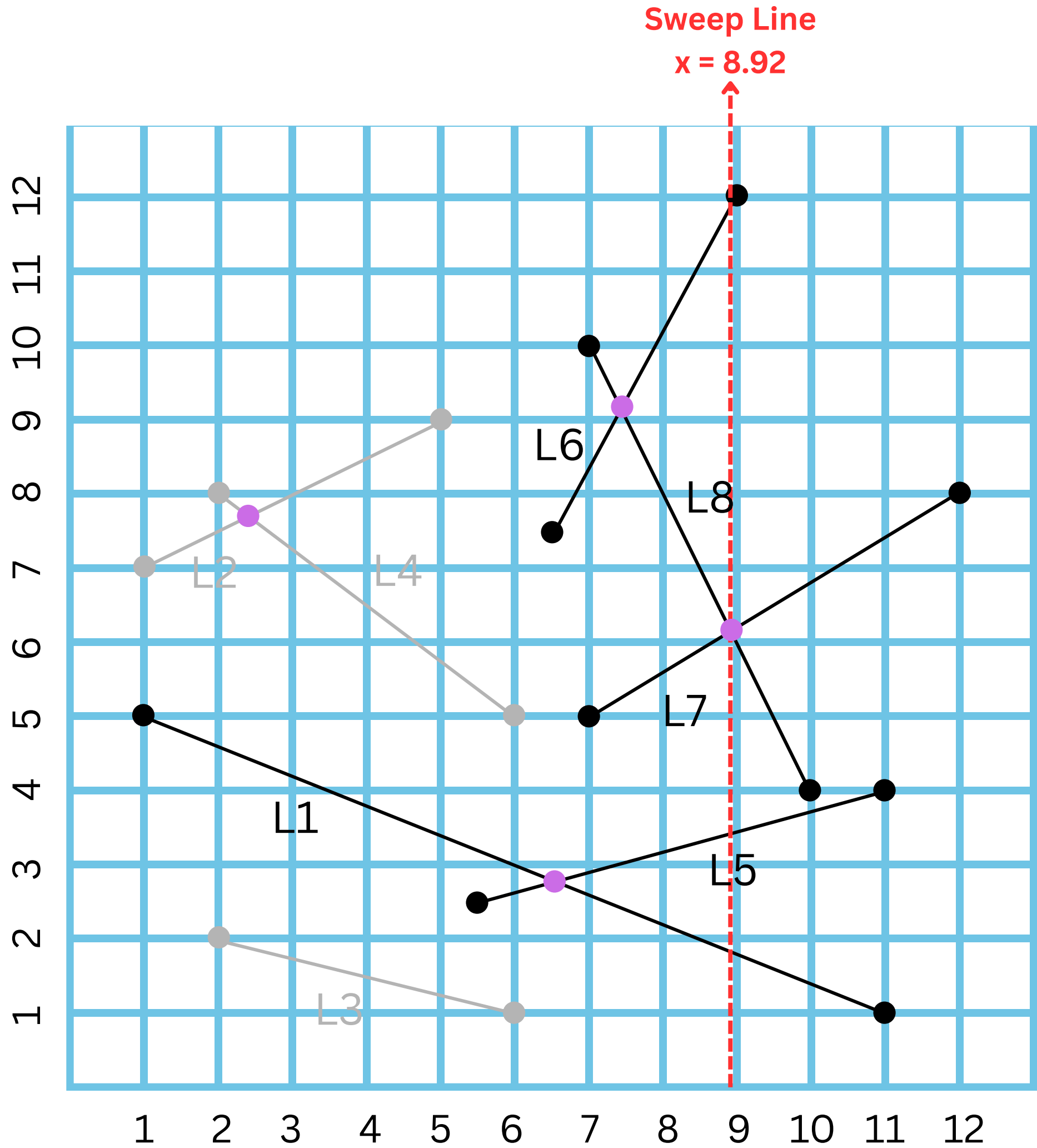
Intersection Test

L8 and L7



x = 8.92

L7e (12, 8)



Event Queue

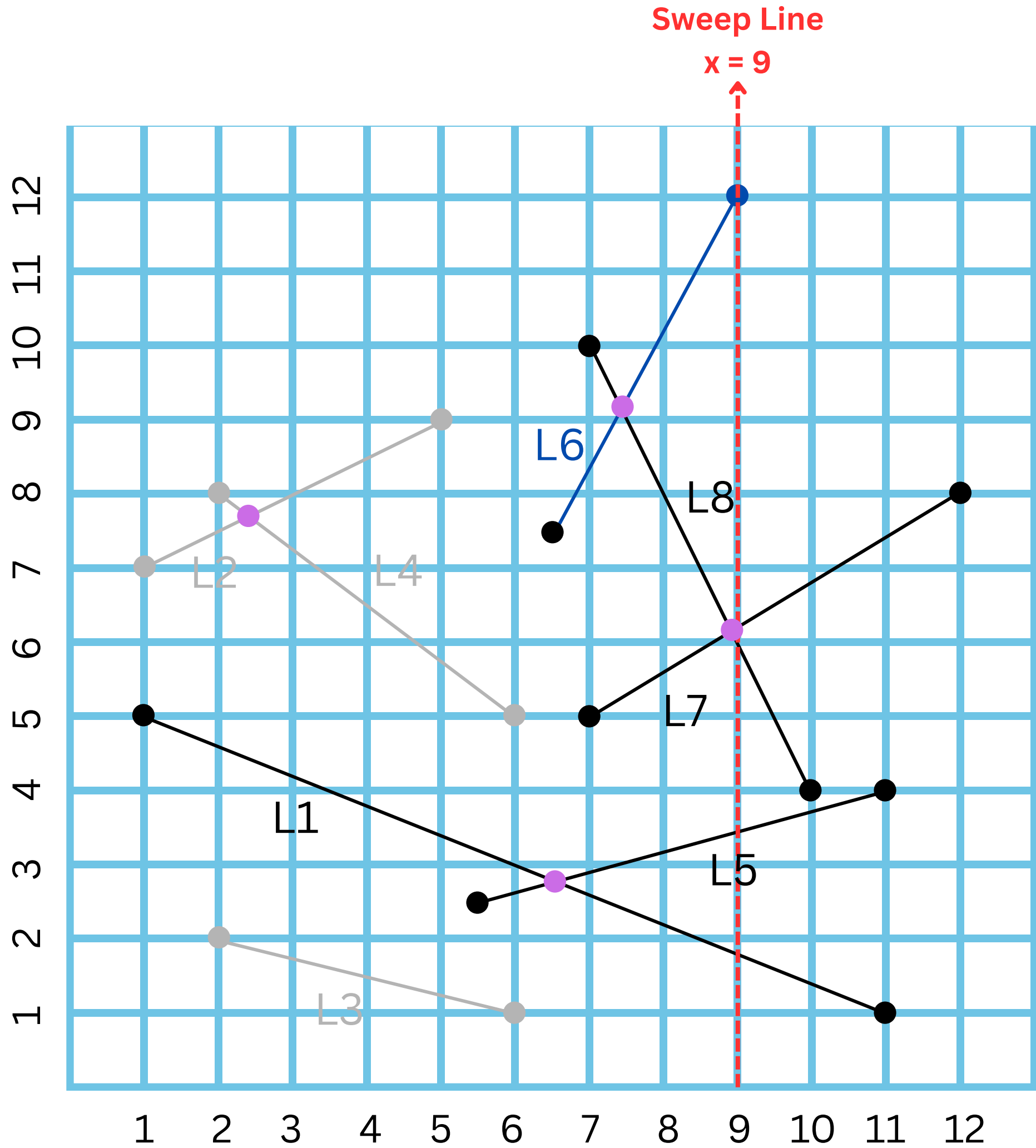
L1b (1, 5)
L2b (1, 7)
L3b (2, 2)
L4b (2, 8)
 $I[L2, L4] (2.4, 7.7)$
L2e (5, 9)
L5b (5.5, 2.5)
L3e (6, 1)
L4e (6, 5)
L6b (6.5, 7.5)
 $I[L1, L5] (6.54, 2.78)$
L7b (7, 5)
L8b (7, 10)
 $I[L8, L6] (7.42, 9.16)$
→ $I[L8, L7] (8.92, 6.15)$
L6e (9, 12)
L8e (10, 4)
L1e (11, 1)
L5e (11, 4)
L7e (12, 8)

Status Tree

L6
L7
L8
L5
L1

Intersection Test

L6 and L7
L8 and L5

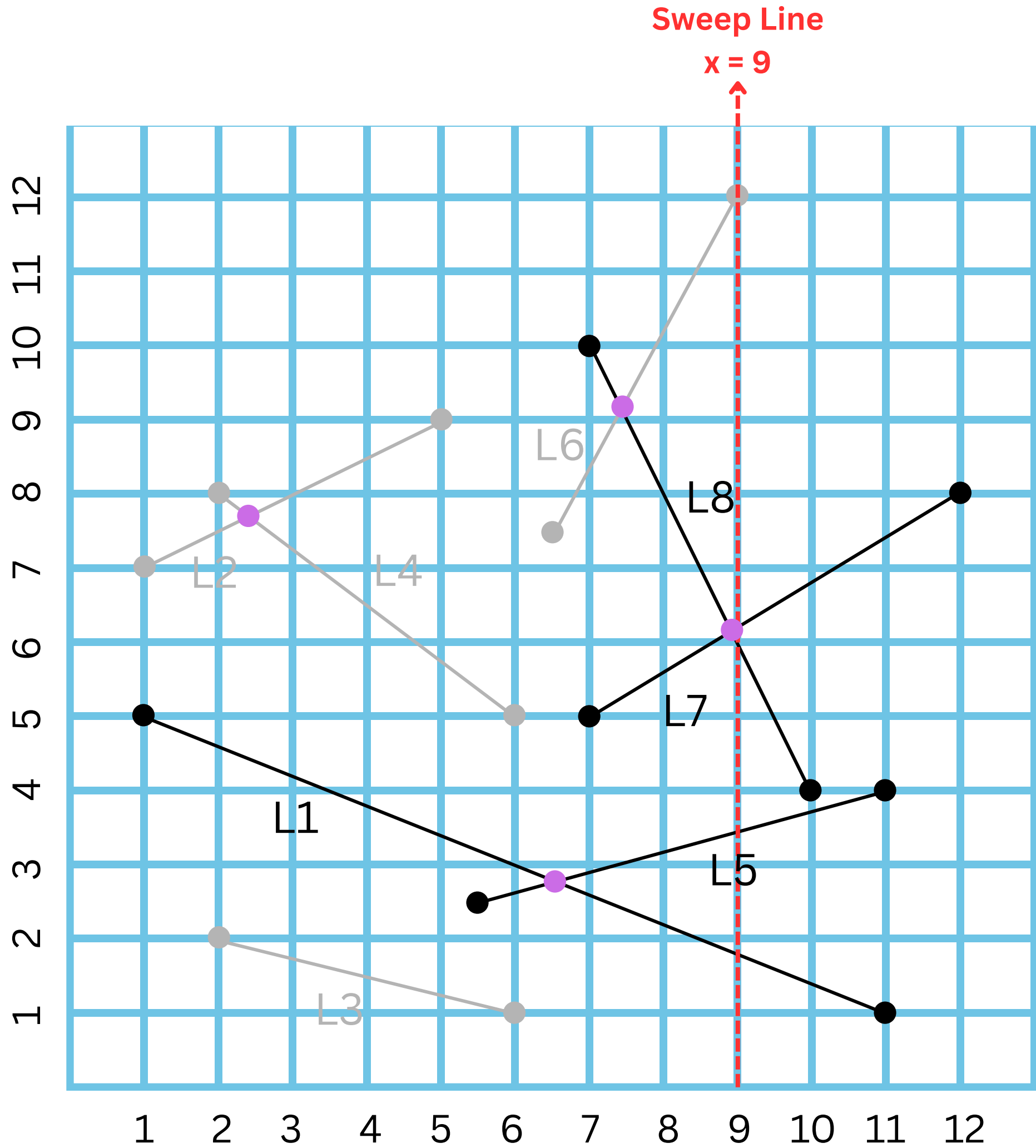


Event Queue

L1*b* (1, 5)
L2*b* (1, 7)
L3*b* (2, 2)
L4*b* (2, 8)
I[**L2**, **L4**] (2.4, 7.7)
L2*e* (5, 9)
L5*b* (5.5, 2.5)
L3*e* (6, 1)
L4*e* (6, 5)
L6*b* (6.5, 7.5)
I[**L1**, **L5**] (6.54, 2.78)
L7*b* (7, 5)
L8*b* (7, 10)
I[**L8**, **L6**] (7.42, 9.16)
I[**L8**, **L7**] (8.92, 6.15)
→ **L6***e* (9, 12)
L8*e* (10, 4)
L1*e* (11, 1)
L5*e* (11, 4)
L7*e* (12, 8)

Status Tree

L6 Delete
L7
L8
L5
L1

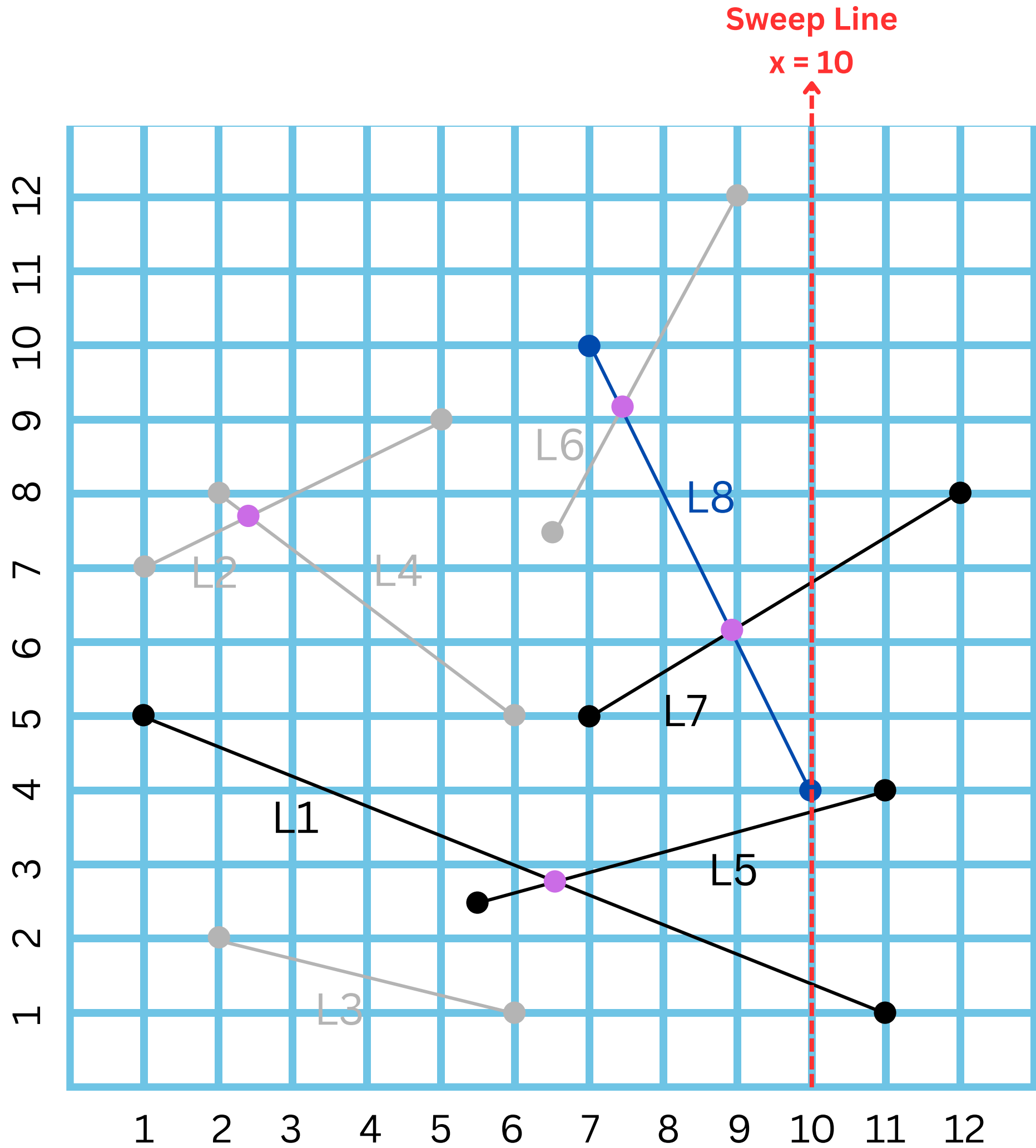


Event Queue

- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7_b (7, 5)
- L8_b (7, 10)
- I[L8, L6] (7.42, 9.16)
- I[L8, L7] (8.92, 6.15)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L7
- L8
- L5
- L1

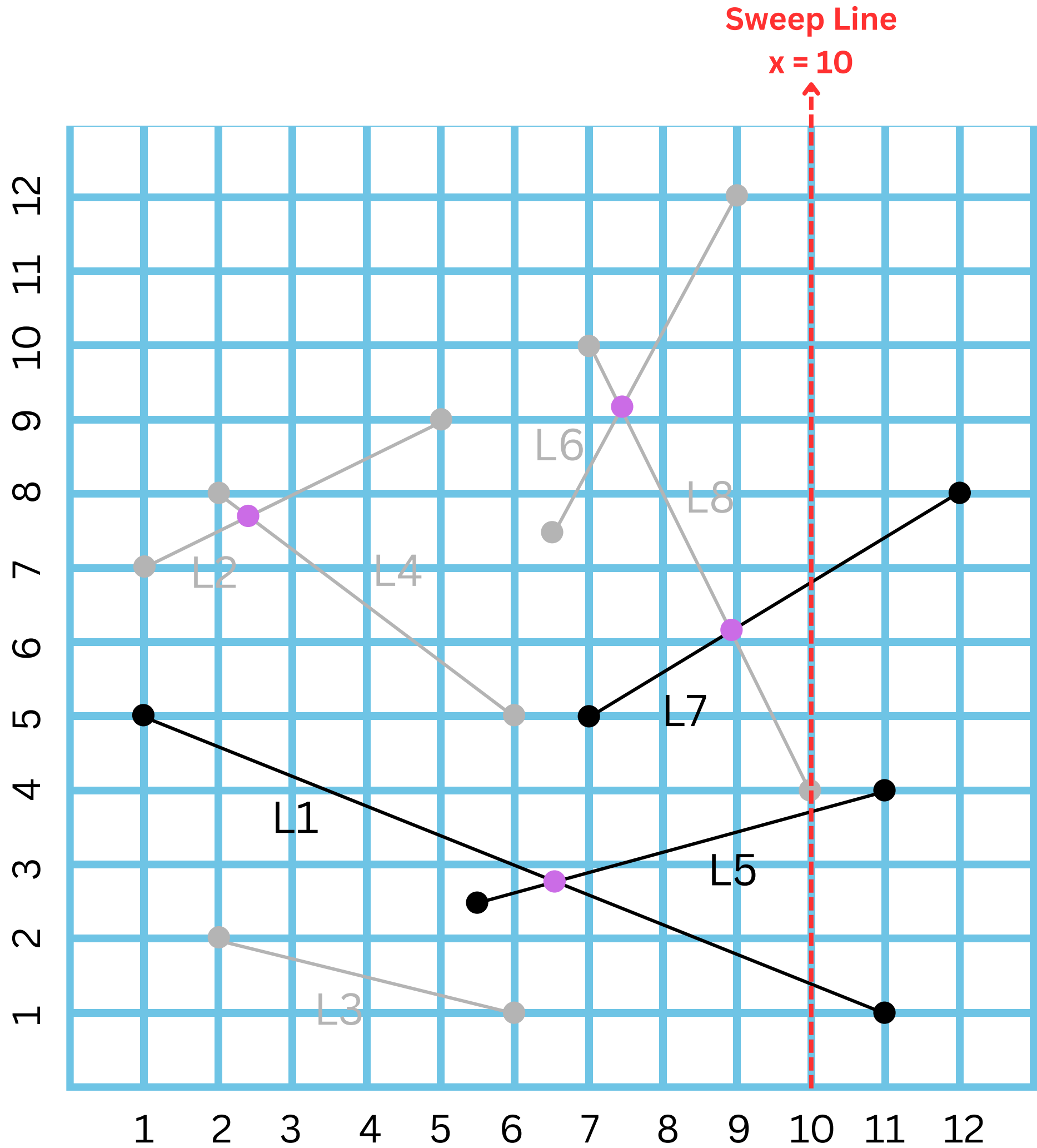


Event Queue

- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7_b (7, 5)
- L8_b (7, 10)
- I[L8, L6] (7.42, 9.16)
- I[L8, L7] (8.92, 6.15)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L7
- L8 Delete
- L5
- L1



Event Queue

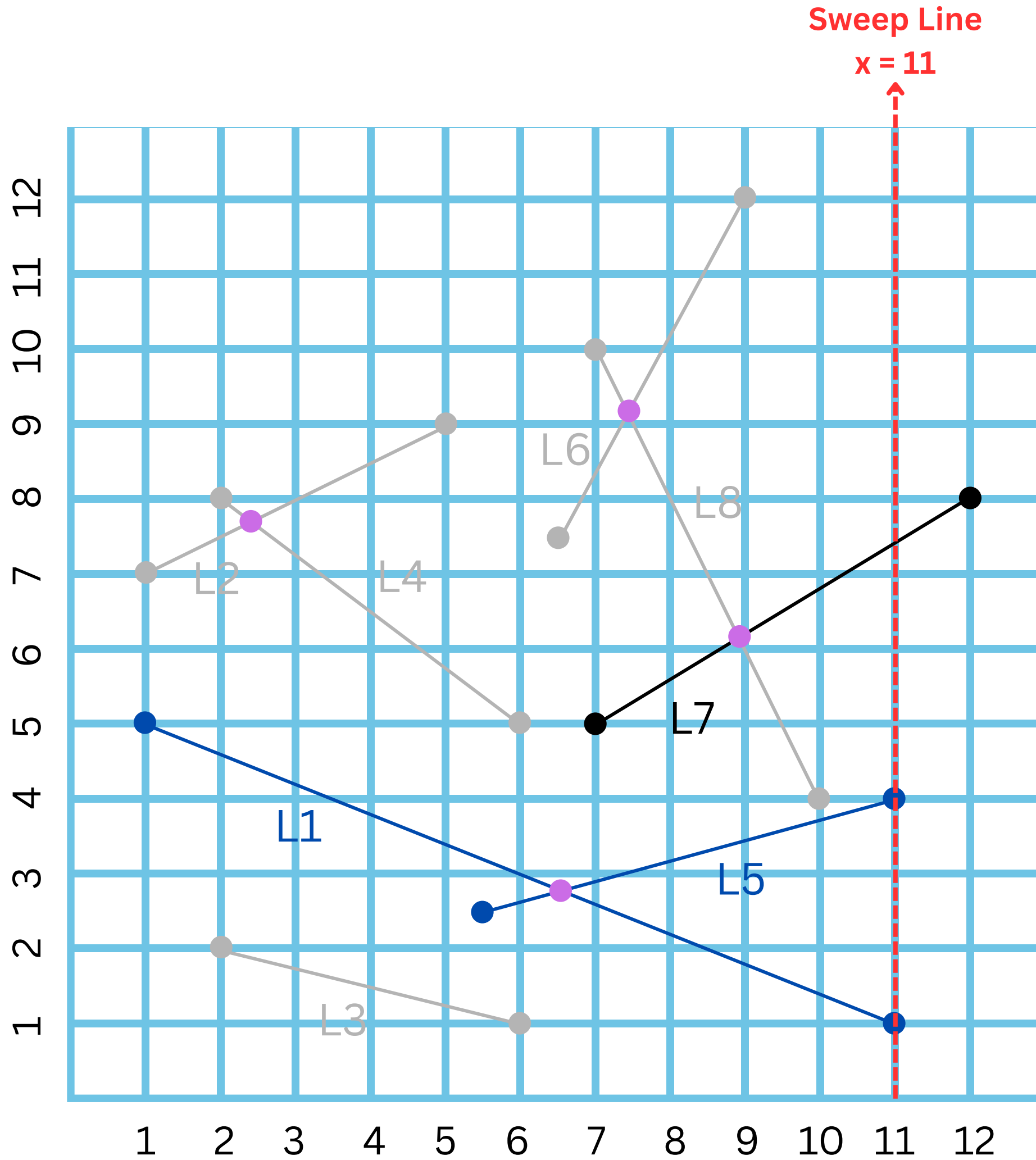
L1*b* (1, 5)
L2*b* (1, 7)
L3*b* (2, 2)
L4*b* (2, 8)
I[**L2**, **L4**] (2.4, 7.7)
L2*e* (5, 9)
L5*b* (5.5, 2.5)
L3*e* (6, 1)
L4*e* (6, 5)
L6*b* (6.5, 7.5)
I[**L1**, **L5**] (6.54, 2.78)
L7*b* (7, 5)
L8*b* (7, 10)
I[**L8**, **L6**] (7.42, 9.16)
I[**L8**, **L7**] (8.92, 6.15)
L6*e* (9, 12)
→ **L8***e* (10, 4)
L1*e* (11, 1)
L5*e* (11, 4)
L7*e* (12, 8)

Status Tree

L7
L5
L1

Intersection Test

L7 and L5

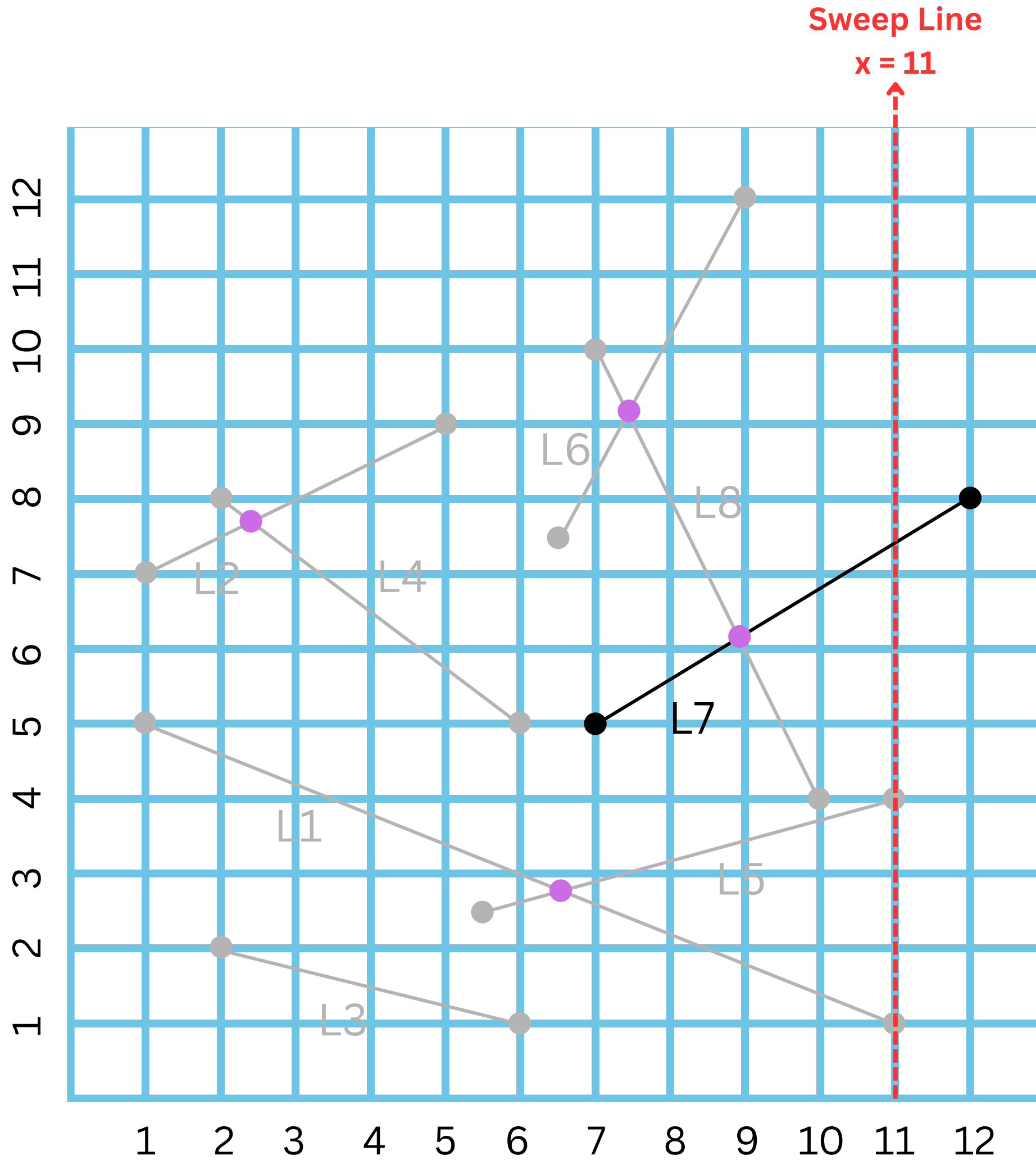


Event Queue

- L1_b (1, 5)
- L2_b (1, 7)
- L3_b (2, 2)
- L4_b (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2_e (5, 9)
- L5_b (5.5, 2.5)
- L3_e (6, 1)
- L4_e (6, 5)
- L6_b (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7_b (7, 5)
- L8_b (7, 10)
- I[L8, L6] (7.42, 9.16)
- I[L8, L7] (8.92, 6.15)
- L6_e (9, 12)
- L8_e (10, 4)
- L1_e (11, 1)
- L5_e (11, 4)
- L7_e (12, 8)

Status Tree

- L7
- L5 Delete
- L1 Delete

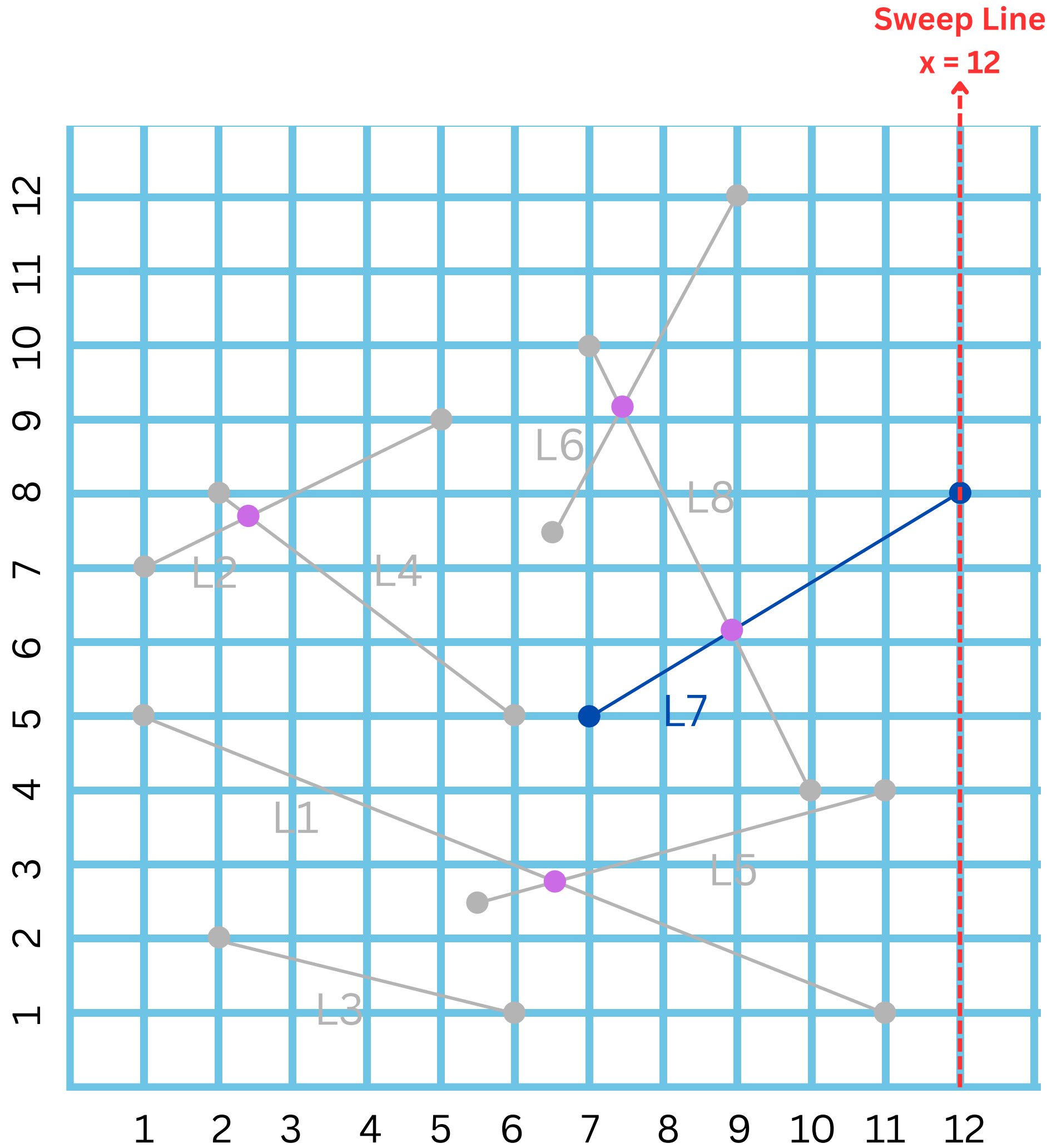


Event Queue

- L1*b* (1, 5)
- L2*b* (1, 7)
- L3*b* (2, 2)
- L4*b* (2, 8)
- I[L2, L4] (2.4, 7.7)
- L2*e* (5, 9)
- L5*b* (5.5, 2.5)
- L3*e* (6, 1)
- L4*e* (6, 5)
- L6*b* (6.5, 7.5)
- I[L1, L5] (6.54, 2.78)
- L7*b* (7, 5)
- L8*b* (7, 10)
- I[L8, L6] (7.42, 9.16)
- I[L8, L7] (8.92, 6.15)
- L6*e* (9, 12)
- L8*e* (10, 4)
- L1*e* (11, 1)
- L5*e* (11, 4)
- L7*e* (12, 8)

Status Tree

L7

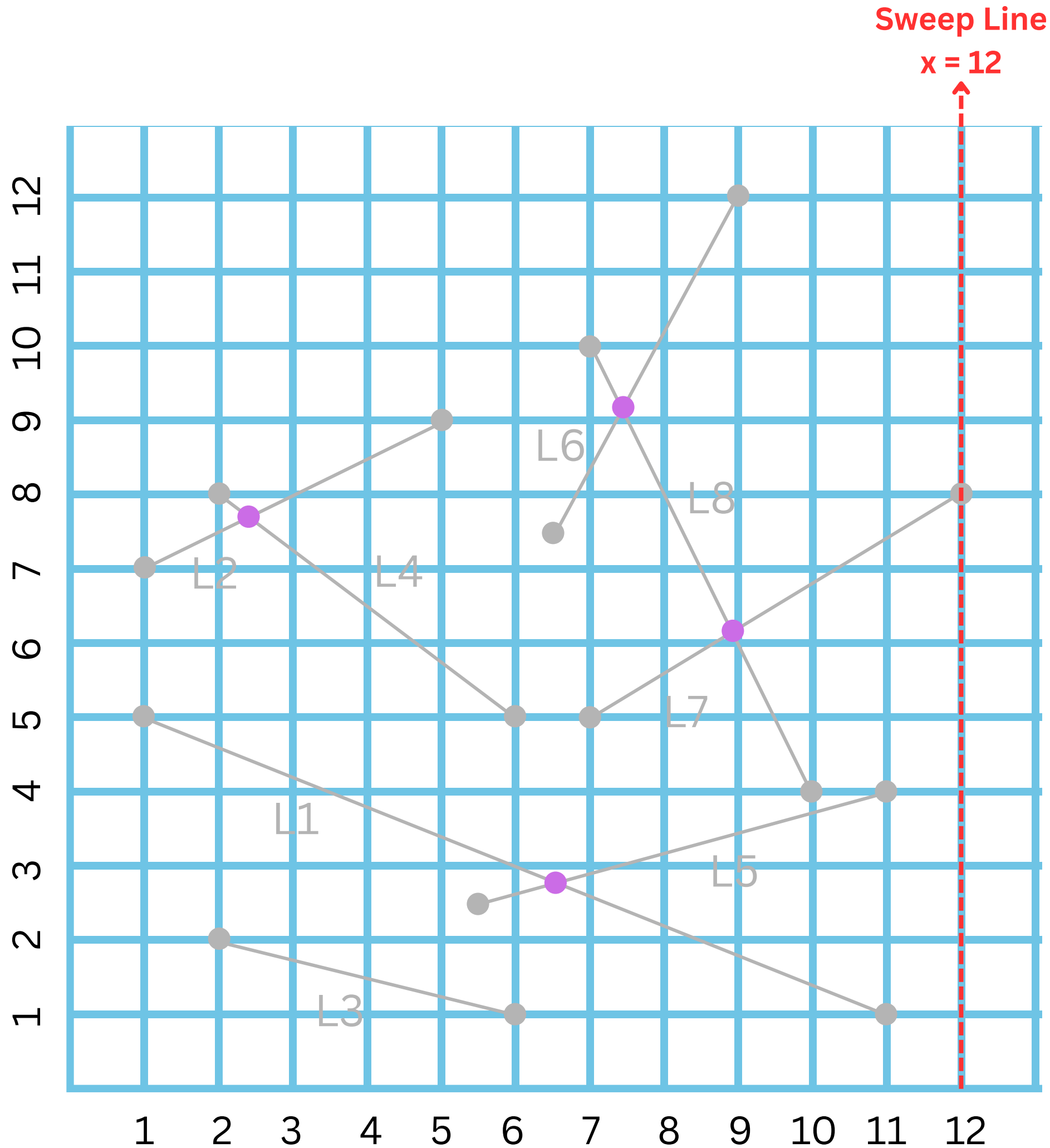


Event Queue

- L1***b* (1, 5)
- L2***b* (1, 7)
- L3***b* (2, 2)
- L4***b* (2, 8)
- I*[**L2**, **L4**] (2.4, 7.7)
- L2***e* (5, 9)
- L5***b* (5.5, 2.5)
- L3***e* (6, 1)
- L4***e* (6, 5)
- L6***b* (6.5, 7.5)
- I*[**L1**, **L5**] (6.54, 2.78)
- L7***b* (7, 5)
- L8***b* (7, 10)
- I*[**L8**, **L6**] (7.42, 9.16)
- I*[**L8**, **L7**] (8.92, 6.15)
- L6***e* (9, 12)
- L8***e* (10, 4)
- L1***e* (11, 1)
- L5***e* (11, 4)
- **L7***e* (12, 8)

Status Tree

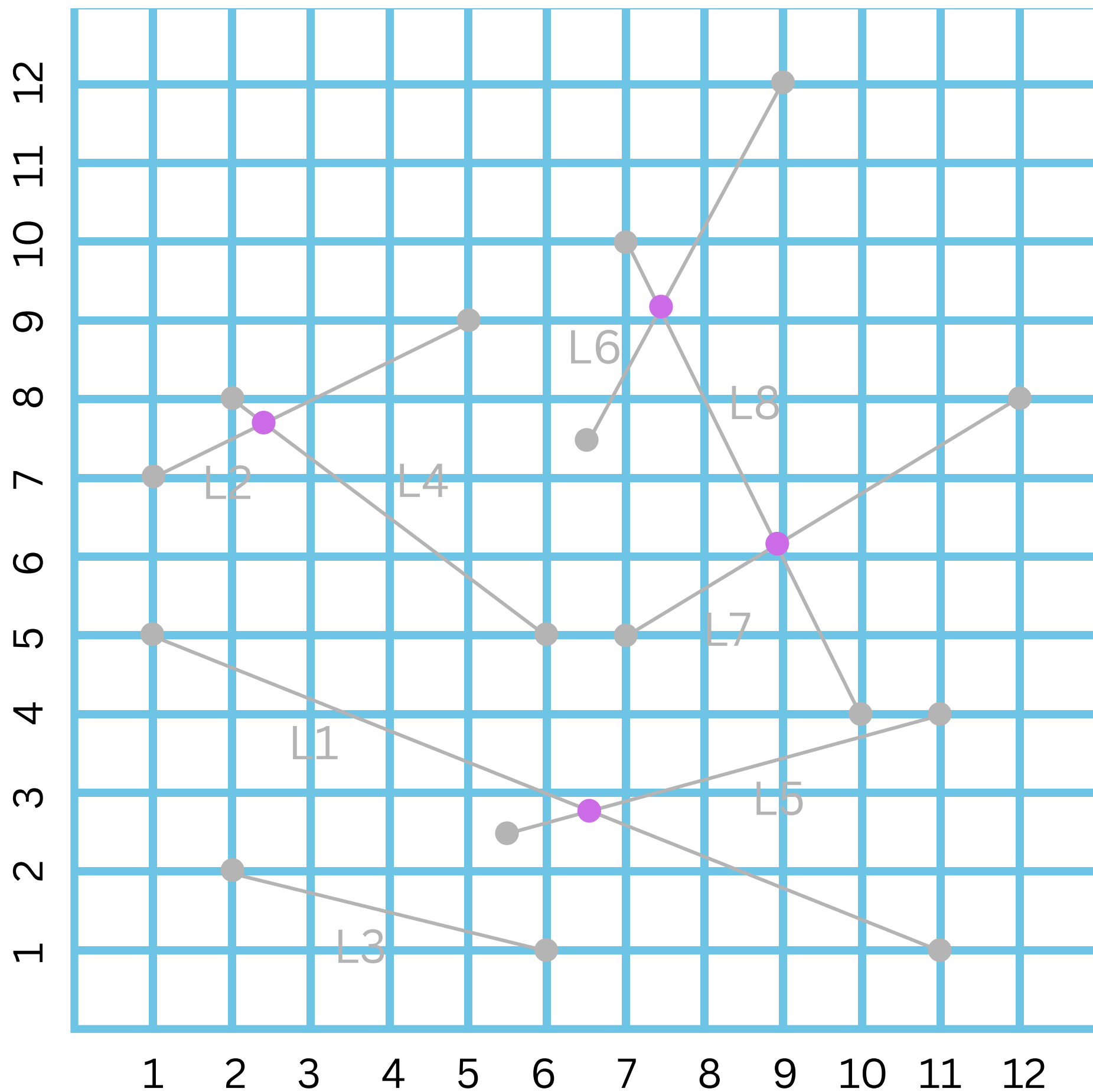
L7 Delete



Event Queue

- L1***b* (1, 5)
- L2***b* (1, 7)
- L3***b* (2, 2)
- L4***b* (2, 8)
- I**[**L2**, **L4**] (2.4, 7.7)
- L2***e* (5, 9)
- L5***b* (5.5, 2.5)
- L3***e* (6, 1)
- L4***e* (6, 5)
- L6***b* (6.5, 7.5)
- I**[**L1**, **L5**] (6.54, 2.78)
- L7***b* (7, 5)
- L8***b* (7, 10)
- I**[**L8**, **L6**] (7.42, 9.16)
- I**[**L8**, **L7**] (8.92, 6.15)
- L6***e* (9, 12)
- L8***e* (10, 4)
- L1***e* (11, 1)
- L5***e* (11, 4)
- L7***e* (12, 8)

Status Tree



Intersection Points

$I[L2, L4]$ (2.4, 7.7)

$I[L1, L5]$ (6.54, 2.78)

$I[L8, L6]$ (7.42, 9.16)

$I[L8, L7]$ (8.92, 6.15)